



Driftwood

Setup & Mod Guide

Minecraft 1.21.1 – NeoForge

Version 3.0.5

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Driftwood — Setup & Mod Guide

Wave -1 — Prerequisites

JDK 21

Tool	Role
Eclipse Temurin JDK 21 (LTS)	Java runtime and development kit for Minecraft 1.21.1

Install the **JDK** (not just JRE) — the JDK includes `javac` and `jlink` needed by NeoForge. Minimum version 21.0.0, any LTS build. On Windows, use the MSI installer and ensure `JAVA_HOME` is set to the install path (e.g., `C:\Program Files\Eclipse Adoptium\jdk-21.0.6.13-hotspot`). On Linux/macOS, use your package manager (`apt install temurin-21-jdk`, `brew install temurin21`).

Verify installation:

```
java -version
# Expected: openjdk version "21.0.x" ...
```

Performance Tuning

Recommended JVM arguments (for XMCL → instance settings → Java → JVM Arguments):

```
-XX:+UseZGC -XX:+ZGenerational -XX:+AlwaysPreTouch -XX:+PerfDisableSharedMem -
XX:SoftMaxHeapSize=12G -Xms8G -Xmx8G
```

System RAM	Recommended allocation
16 GB	<code>-Xms6G -Xmx10G -XX:SoftMaxHeapSize=8G</code>
32 GB	<code>-Xms12G -Xmx22G -XX:SoftMaxHeapSize=12G</code>
64 GB	<code>-Xms12G -Xmx32G -XX:SoftMaxHeapSize=12G</code>

Note: ZGC (Z Garbage Collector) with generational mode is the recommended GC for Java 21 with NeoForge. It provides consistently low latency (sub-millisecond pause times) and handles the large heap sizes common with modded Minecraft better than Shenandoah or G1GC. The Adoptium Temurin JDK 21 includes ZGC — no special Java build needed. `-XX:+AlwaysPreTouch` pre-initializes all memory pages at startup, preventing runtime page-fault stalls. `-XX:+PerfDisableSharedMem` disables synchronous `hsperfdata` file writes, removing a minor source of I/O-induced latency. `-XX:SoftMaxHeapSize=12G` tells ZGC to target ~12G heap usage and GC more aggressively to stay near that level, while allowing bursts above it up to `-Xmx`. For this to take effect, `-Xmx` must be set higher than `-Xms`. The baseline uses equal values for simplicity; on a 64 GB machine you may want e.g. `-Xms12G -Xmx24G` instead to let `SoftMaxHeapSize` work.

X Minecraft Launcher (XMCL)

Tool	Role
XMCL	Mod manager and launcher for the modpack

XMCL manages mod and modpack installation, updates, and launching for this modpack. It supports both **CurseForge** and **Modrinth** as mod sources in one launcher, handles NeoForge

installation automatically, manages Java versions per instance, and uses hard links to avoid duplicate mod storage. Open source, free, with no premium tier.

Setup:

1. Download and install **XMCL** from xmcl.app for your OS.
 - Windows: App Installer, AppX, or Zip (x64)
 - macOS: DMG (Intel or Apple Silicon)
 - Linux: Deb, RPM, AppImage, Flathub
 - Or via winget: `winget install CI010.XMinecraftLauncher`
2. Create a new instance: click + → select Minecraft **1.21.1** and modloader **NeoForge** (latest recommended for 1.21.1).
3. XMCL downloads NeoForge and the correct Java version automatically.
4. Launch the instance once to generate the `mods/` directory and configs.
5. Verify the main menu shows “NeoForge X.X.X” in the bottom-left corner.

From there, install mods through the instance’s **Mods** tab from either CurseForge or Modrinth, or by dropping `.jar` files into the instance’s `mods/` folder. XMCL uses hard links — mods are stored once globally and linked per instance, avoiding duplicates. All mod links in this document link to their CurseForge pages — XMCL can install from CurseForge directly via the Mods tab.

For texture packs, drop `.zip` files into the instance’s **Resource Packs** tab or folder. For shaderpacks, drop `.zip` files into the instance’s **Shader Packs** tab or folder. Access the instance folder from the instance settings.

Minecraft 1.21.1

Install the **Minecraft 1.21.1** vanilla client from the official launcher:

1. Open the [Minecraft Launcher](#).
2. Create a new installation with version **release 1.21.1**.
3. Run it once to generate the vanilla `versions/` and `libraries/` directories, then close.
4. Locate your Minecraft directory:
 - Windows: `%APPDATA%\minecraft`
 - Linux: `~/.minecraft`
 - macOS: `~/Library/Application Support/minecraft`

NeoForge

Tool	Role
NeoForge	Mod loader for Minecraft 1.21.1

NeoForge is the mod loader powering this modpack. If you’re using **XMCL**, it installs NeoForge automatically when you create a NeoForge instance (see [XMCL](#) above) — skip to step 5 to verify.

If installing manually:

1. Download the **NeoForge installer** for Minecraft 1.21.1 from neoforged.net.
2. Run the installer: `java -jar neoforge-1.21.1-<version>-installer.jar`.
3. Select “**Install client**” and point it to your Minecraft directory (auto-detected on most systems).
4. Launch the new **neoforge** profile in the Minecraft Launcher once to generate the `mods/` directory and NeoForge configs.
5. Verify the main menu shows “NeoForge X.X.X” in the bottom-left corner.

Important: Do not add any mod `.jar` files yet. Wave -1 is complete when the vanilla 1.21.1 client runs under NeoForge with no mods installed.

Skip this if you've played Minecraft before.

Controls (Default)

Action	Key
Move	W A S D
Jump	Space
Sneak	Shift
Sprint	Ctrl
Attack / Break	Left-click
Use / Place	Right-click
Inventory	E
Drop	Q
Hotbar slots	1–9

Your First 10 Minutes

1. **Punch a tree** — Hold left-click on a tree trunk until it breaks. Get 8+ logs.
2. **Make a crafting table** — Open inventory (E), place 1 log → 4 planks. Fill all 4 crafting slots with planks → crafting table.
3. **Craft tools** at the crafting table (3×3 grid):

Tool	Pattern
Wooden Pickaxe	3 planks top row, 2 sticks middle column
Wooden Axe	3 planks top-left, 2 sticks down
Wooden Sword	2 planks middle column, 1 stick below

1. **Gather stone** — Find grey stone, mine 8–12 blocks with your pickaxe.
2. **Build a shelter** — Dirt/wood walls + door (6 planks, 2 columns of 3) + torches (stick + coal/charcoal).
3. **Craft a furnace** — 8 cobblestone in a ring (center empty). Use it to smelt ore, cook meat, make charcoal.
4. **Survive the night** — Stay inside. Dig, expand, or sleep if you have a bed (3 wool + 3 planks).
5. **Day 2 priorities:** Find food, get iron (dig staircase-down until you see brown speckled blocks), find a village, make a bed.

Keybind to Set Immediately

Open **Options** → **Controls** → **Key Binds** and set these before anything else:

Action	Suggested Key	Why
Open Waystone Menu	Ctrl+Shift+W	Fast travel anywhere
Zoom	C	Sodium zoom
Toggle Dynamic Lights	Ctrl+L	Torch in off-hand = mobile light
Open Backpack	B	Sophisticated Backpacks
Open Fullscreen Map	J	JourneyMap

Wave 0 — Foundation

The essentials that make the game run well. Dependencies, performance, rendering, shaders, chunk loading, and texture packs. Install everything in this wave before launching.

Performance & Rendering

Mod	Role
Sodium	Rendering engine
Iris	Shader loader (v1.8.14, NeoForge native)
Lithium	Server-side game logic optimization
Distant Horizons	LOD rendering
ModernFix	All-in-one perf + bug fixes
ImmediatelyFast	Immediate-mode rendering
Entity Culling	Hide off-screen entities
Dynamic FPS	Background FPS reduction
NoisiumForked	Worldgen optimization (active fork)
FerriteCore	Memory usage reduction

ModernFix (configurable in-game via Mod List → ModernFix → Config, or `config/modernfix-mixins.properties`):

- `mixin.perf.dynamic_resources = true` — Load block/item models lazily instead of all at startup. **Biggest single launch-time speedup** in the pack. Reduces memory usage too. Disable if you see missing textures or broken models.

Shaderpack

Shaderpack	Description
Complementary Unbound	Realistic visual style, Potato→Ultra profiles, block-specific effects
Euphoria Patches	Optional add-on for Complementary Shaders — extends with handcrafted features, all disabled by default

Euphoria Patches is a shader add-on, not a standalone shaderpack. It patches Complementary Shaders (Reimagined or Unbound) with additional visual options. Install via its patcher mod (requires NeoForge, no additional mod dependencies). All features are disabled by default and can be individually enabled from the shader options menu.

Settings for NVIDIA RTX 4080 SUPER (16GB VRAM, 4K) — target: stable 70-80 FPS:

Setting	Value
Visual Style	Unbound
Profile	High
RP Support	Integrated PBR+
Shadow Resolution	2048
Shadow Distance	12 chunks
Real-Time Shadows	High
Light Shaft Quality	Medium

Setting	Value
Entity Shadows	OFF
Detail Quality	High
Cloud Quality	High
Water Reflection Quality	Medium (max)
Block Reflection Quality	Medium
Anti-Aliasing (FXAA)	ON
Edge Shadow (SSAO) Quality	Medium
Advanced Colored Lighting	8 chunks
Texture Filtering	8x
Bloom	OFF

General MC settings: Graphics **Fancy**, Render Distance **10 chunks**, Simulation Distance **8 chunks**, Biome Blend **2 blocks**, Mipmap Levels **4x**, Entity Distance **100%**, Fullscreen ON, VSync OFF, Max Framerate **120**. Always Defer Chunk Updates **Enabled**, Use No Error Context **Enabled**.

Distant Horizons handles far rendering — keep vanilla render distance low. DH LODs render through the shader pipeline; start LOD distance at **128** and push to **256** if the FPS budget allows. Monitor VRAM usage (shaders + DH + high-res texture packs can peak past 12GB at 4K). If you experience dips below 70 FPS, first reduce Shadow Resolution to **1024**, then lower Shadow Distance to **8 chunks**.

Texture Packs

Pack	Role
Fresh Animations	Living entity animation overhaul — mobs blink, look around, show emotion
Alacrity	Full 32x conversion — RPG-themed textures, GUI overhaul, custom 3D models
Alacrity — Just Enough Items Support	Retextures JEI (+8 addons) to match Alacrity's RPG style
Alacrity — Shulker Box Tooltip Support	Retextures Shulker Box Tooltip previews to match Alacrity's RPG style
Enhanced Boss Bars	Redesigned boss bars with themed designs for vanilla and modded bosses
RAY's 3D Rails	3D block & item textures for all rail types
Better Lanterns	Enhanced 3D look for lanterns and chains
Smooth Font 32x	Smooth 32x font replacement for cleaner text readability

Fresh Animations adds idle animations, directional looking, sleep/blink cycles, and emotional states to all vanilla mobs. It's a resource pack overlay (not a mod) — place it high in the pack order. Works on any version with no mod dependencies.

Alacrity is a full 32x conversion with an RPG-inspired art style — all blocks, items, GUI elements, and mobs are redesigned with a cohesive fantasy look. Draws from Warcraft, Elder Scrolls, and Gothic for its aesthetic. Custom 3D models (CEM) require a mod like Entity Model Features but the base pack works with any setup.

Alacrity Mod Support packs (by TheDatThai) extend Alacrity's RPG styling to mod UIs: JEI and Shulker Box Tooltip. Place these **below** the base Alacrity pack — putting them above corrupts JEI/Shulker graphics (mismatched textures).

Enhanced Boss Bars replaces boss bar textures with themed designs — covers vanilla bosses and has support for Twilight Forest bosses. For full compatibility with L_Ender's Cataclysm and The Aether (which use custom boss bar rendering), install the companion [Enhanced Boss Bars mod](#). Pure resource pack for vanilla/Twilight Forest bars.

RAY's 3D Rails replaces rail block and item textures with 3D models. Supports shaders since v3.0. Compatible with all rail types from Steam 'n' Rails and Create.

3D Ladders replaces the vanilla ladder model with a 3D version. Simple, lightweight, no conflicts.

Better Lanterns gives lanterns and chains an enhanced 3D look with connected chain models. Compatible with Arcane Lanterns and Thin Air mods.

Installation: Install via XMCL — drop [.zip](#) files into the instance's Resource Packs tab, or copy them into the instance's [resourcepacks/](#) folder (access via instance settings). Load order (top = highest priority):

1. Fresh Animations
2. Alacrity
3. Alacrity — Just Enough Items Support
4. Alacrity — Shulker Box Tooltip Support
5. Enhanced Boss Bars
6. RAY's 3D Rails
7. 3D Ladders
8. Better Lanterns
9. Smooth Font 32x

Infrastructure

Mod	Role
Chunk Loaders	Keep chunks loaded across dimensions — essential for Phase 2+ automation
SuperMartijn642's Config Lib	Config library (Chunk Loaders dependency)
SuperMartijn642's Core Lib	Core library (Chunk Loaders dependency)

The essentials that make the game run well. Dependencies, performance, rendering, shaders, chunk loading, and texture packs. Install everything here before launching.

Install

Use **XMCL** — create a 1.21.1 NeoForge instance, install all Wave 0 mods via the Mods tab. See [MODLIST.md](#) for the full list.

Wave 0.5 (Player Experience) can be installed at the same time — they're independent mods that don't conflict.

First Launch Checklist

1. Video Settings

Setting		Recommended	Notes
Graphics		Fabulous	Sodium makes this fast
Render Distance		10 chunks	Distant Horizons handles the rest
Simulation Distance		8 chunks	Fewer distant mobs ticking = better FPS
VSync		OFF	
Max Framerate		Uncapped	

2. Distant Horizons (LOD Mod)

Open **Video Settings** → [colored tiles icon] next to the FOV slider.

Setting		Value
Enable Rendering		ON
Enable Distant Generation		ON
LOD Render Distance Radius		128–256
Quality Preset		Medium
CPU Load		Aggressive
Enable Cloud Rendering		OFF
Show LOD Gen Progress		ON

Tip: DH generates LOD data as you explore. First visit to an area has temporary pop-in. Terralith + Tectonic terrain takes longer initially. Monitor VRAM — shaders + DH at 4K can push past 12GB.

3. Shader Setup

Drop **Complementary Unbound .zip** into [shaderpacks/](#). In-game: Options → Video Settings → Shaderpacks → select Complementary Unbound. Works with Iris on NeoForge.

Euphoria Patches (optional add-on): Install the patcher mod, it auto-detects Complementary and applies additional visual options. All features disabled by default.

Settings for RTX 4080 SUPER (16GB VRAM, 4K):

- Visual Style: Unbound, Profile: High, RP Support: Integrated PBR+
- Shadow Resolution: 2048, Shadow Distance: 12 chunks
- Real-Time Shadows: High, Light Shaft Quality: Medium
- Entity Shadows: OFF, Cloud Quality: High, Water Reflection: Medium
- Bloom: OFF

General MC: Graphics Fancy, Render Distance 10, Simulation 8, Biome Blend 2, Mipmap 4x, Entity Distance 100%, VSync OFF, Max Framerate 120.

4. Texture Pack Load Order (top = highest priority)

1. Fresh Animations
2. Alacrity
3. Alacrity — JEI Support

4. Alacrity — Shulker Box Tooltip Support
5. Enhanced Boss Bars
6. RAY's 3D Rails
7. 3D Ladders
8. Better Lanterns
9. Smooth Font 32x

Alacrity support packs go BELOW the base Alacrity pack — putting them above corrupts JEI graphics.

5. JVM Arguments (XMCL → instance settings → Java → JVM Arguments)

```
-XX:+UseZGC -XX:+ZGenerational -XX:+AlwaysPreTouch -XX:+PerfDisableSharedMem -
XX:SoftMaxHeapSize=12G -Xms8G -Xmx8G
```

System RAM	Allocation
16 GB	<code>-Xms6G -Xmx10G -XX:SoftMaxHeapSize=8G</code>
32 GB	<code>-Xms12G -Xmx22G -XX:SoftMaxHeapSize=12G</code>
64 GB	<code>-Xms12G -Xmx32G -XX:SoftMaxHeapSize=12G</code>

Performance Mods — Install and Forget

These mods optimize automatically. No config needed.

Mod	What it does
Sodium	Modern rendering engine — massive FPS boost
Iris	Shader loader (Complementary Unbound)
Lithium	Server-side logic optimization — faster AI, block updates
Distant Horizons	LOD rendering — see 256+ chunks away
ModernFix	All-in-one performance + bug fixes
ImmediatelyFast	Immediate-mode rendering for blocks/items
Entity Culling	Skip rendering off-screen entities
Dynamic FPS	Lower FPS when Minecraft is backgrounded
NoisiumForked	Faster world generation when exploring
FerriteCore	Lower memory usage — fewer OOM crashes

ModernFix critical setting: `mixin.perf.dynamic_resources = true` in `config/modernfix-mixins.properties` — biggest single launch-time speedup. Disable only if you see missing textures.

Infrastructure

Chunk Loaders

Keeps chunks loaded when you're far away or in another dimension.

Type		Coverage		Use	
Single Chunk Loader		1 chunk		One machine or farm	
Spotloader		Handheld, temporary		Explore nearby	
Multi Chunk Loader		3×3 chunks		Factory complex or colony	

How: Craft (check JEI — iron + redstone + ender pearl), place, right-click to toggle. Colored border shows loaded area.

Placement priority (late game):

1. Main base (ME drive, Mekanism induction matrix, Create power core) — Multi Chunk
2. MineColonies colony — Single or Multi
3. Mining outposts (Digital Miner) — Single
4. Other dimensions — Single per outpost

No performance concern with a dozen loaders on a modern CPU.

Wave 0.5 — Player Experience

All the visual polish, UI improvements, inventory tools, storage, travel, trading, and quality-of-life features that make the game feel great.

Visual & Client QoL

Mod	Role
LambDynamicLights	Dynamic lighting
BetterF3	Customizable debug HUD
Sound Physics Remastered	Realistic sound
AmbientSounds	Nature ambiance (birds, wind, water)
The Immersive Music Mod	147 new biome/structure-based songs, vanilla silence gaps preserved
Immersive UI	Animated UI, smooth hotbar, item particles
Status Effect Bars Reforged	Duration bars on status effect icons in HUD and inventory
Traveler's Titles	RPG-style titles entering biomes/dimensions
Better Modlist	Enhanced mod list screen with badges/categories
Better Third Person	Independent 360° camera rotation in third-person view
Tooltip Overhaul	Modern, sharp tooltip rendering with equipment compare
First Person Model	Full body rendering in first-person view (requires Not Enough Animations)
3D Skin Layers	Renders player skin layers (jackets, hats) with 3D depth
Presence Footsteps	Block-aware footstep sounds for every surface type
Particular Reforged	Ambient particles: 3D water splashes, fireflies, falling leaves, cave dust, waterfall cascades
Wakes Reforged	Realistic water wakes and splashes when traveling on water
Sounds	170+ new sound effects for UIs, items, blocks, actions
Better Days	Customizable day/night cycle length, enhanced sleep
Better Safe Bed	Prevents mobs that can't reach you from blocking sleep — chill-rule essential
Beautiful Enchanted Books [Mod Edition]	Unique textures per enchantment book type — standalone mod, no resource pack needed
Overflowing Bars	Expanded health, armor, and toughness bars beyond vanilla limits
Cherished Worlds	Favorite/pin worlds to top of list, prevent accidental deletion
Day Counter	Day counter title overlay with milestone celebrations — appears at dawn each day

Mod	Role
Eating Animation	First-person eating/drinking animation — food moves to mouth and shrinks
Camera Overhaul	Camera tilting, screen shakes, mouselook roll, idle sway — satisfying movement feedback
Dynamic Crosshair	Context-aware crosshair — hides/shows/changes style based on what you're looking at or holding
[DLS] Dark Loading Screen	Replaces the default bright-red loading screen with a dark version — easier on the eyes

Inventory & UI

Mod	Role
JEI	Recipe viewer
Jade	Block info HUD
Jade Addons (Neo/Forge)	Extra Jade info panels
JourneyMap	Real-time minimap + full-screen world map, waypoints, entity radar
JourneyMap Integration	Shows Waystones as icons on JourneyMap
AppleSkin	Food hunger/saturation overlay
Mouse Tweaks	Inventory management shortcuts
Inventory Essentials	Drag-to-transfer, single-item moves, sorting
Fancy Toasts	Beautiful animated advancement popups
Shulker Box Tooltip	Preview shulker box contents on hover
Item Borders	Colored borders on items based on rarity
Loot Journal	Animated item pickup notifications
Reliable Advancements	Overhauled advancements UI with editor/pan/zoom
Polymorph	Choose crafting result when recipes conflict

Storage & Travel

Mod	Role
Sophisticated Storage	Upgradable barrels/chests
Sophisticated Backpacks	Portable inventory
Sophisticated Storage Create Integration	Full-featured Sophisticated Storage on Create contraptions
Sophisticated Backpacks Create Integration	Full-featured Sophisticated Backpacks on Create contraptions
Waystones	Fast travel between waypoints
GraveStone Mod	Keep inventory on death

Utility & QoL

Mod	Role
Easy Anvils	Overhauled anvils — persistent items, reduced prior work penalties, free renaming, repair with blocks

Easy Anvils requires **Puzzles Lib** — already in Wave 5 as a Mutant Monsters dependency. No additional mods needed.

Village & Trading

Mod	Role
Dynamic Villager Trades Neoforge 1.0.0	Dynamically changing villager trades — villagers specialize in what you buy from them, reducing the need for massive trading halls
Improved Village Placement	Forces villages to spawn on flat terrain — prevents cliff-side and underwater villages

Configuration — Village & Trading

Dynamic Villager Trades (`config/dynamicvillagertraders.properties`):

- `global_randomness = 0.7` (recommended, default 1.0) — Lower values make villagers settle on specific trades faster; higher values increase time to get desired trades. Multiplied by group randomness in the datapack.
- `refresh_delay = 1` (recommended, default 0) — Number of restocks between trade changes. `1` means trades change every other restock, giving you time to use existing trades before they rotate. Trades always change on level-up regardless.
- `no_book_duplicates = true` — Prevents librarians from selling the same enchant at different levels simultaneously.
- `enchant_repair_compat = false` — Adds Curse of Vanishing to every book sold by librarians (for Enchant & Repair compatibility). Keep disabled unless that mod is added.

Configuration — Wave 0.5

Waystones (`config/waystones-common.toml`):

- `inventoryButton = "any"` — Opens waystone menu from inventory without a keybind.
- `enableCosts = false` — Free teleportation.

Loot Journal (`config/obscuria/loot_journal-client.toml`):

- `trackItemPickups = true` — Logs all item pickups for per-session statistics.
- `trackXpPickups = true` — Logs all XP pickups for per-session statistics.

Better Days (`config/betterdays-common.toml`):

Time speed can be set via three modes (`speedMethod`):

Mode	How it works	Best for
MINUTES (default)	Set <code>daySpeedMinutes</code> and <code>nightSpeedMinutes</code> in real minutes	Simple: 10 min day / 10 min night
RATIO	Multiplier relative to vanilla (1.0 = vanilla 20min cycle)	Fine control: 0.5 = 40 min, 2.0 = 10 min

Mode	How it works	Best for
SEASON	Day length varies by season using <code>seasonDayMinutes</code> and <code>seasonLatitude</code>	Immersive — day shortens in winter, lengthens in summer

Recommended settings for this pack (`speedMethod = "MINUTES"`):

- `daySpeedMinutes = 20` — Days last 20 minutes (2x vanilla). Gives relaxed building time.
- `nightSpeedMinutes = 10` — Nights remain 10 minutes. Sleep acceleration keeps them short.
- `enableSleepFeature = true` — Smooth time acceleration while in bed.
- `dayStart = 0` — Dawn at time 0. Night starts at 13000 (default, no change needed).
- `nightStart = 13000` — Dusk starts at time 13000 (default).

Time effects — all disabled by default. Leave them off:

- `weatherEffect = "SLEEPING"` — Lets rain/storms finish faster when sleeping (safe — NeoForge handles this natively anyway).
- `potionEffect = "NEVER"` — Keep off. Speeding potion timers while sleeping is confusing.
- `hungerEffect = "NEVER"` — Keep off. Speeding hunger while sleeping is annoying.
- `blockEntityEffect = "NEVER"` — Keep off. Enabling this accelerates furnaces/hoppers during sleep, which can desync items and wastes fuel.
- `cropEffect = "NEVER"` — Keep off. Conflicts with Serene Seasons crop cycle timers.

Regarding the SEASON mode: Since this pack includes **Serene Seasons**, you may want to try `speedMethod = "SEASON"` (*not* "SEASONS" — the value is singular) for a more immersive cycle where summer days are long and winter days are short. However, this is experimental — Serene Seasons already handles seasonal crop growth and temperature; Better Days would only change day length visually. Test before committing.

The `seasonDayMinutes` and `seasonLatitude` fields exist in the default config but may not appear in the file until `speedMethod = "SEASON"` is set and the game reloads. If they're missing, add these lines manually under the `[time]` section:

```
speedMethod = "SEASON"
seasonDayMinutes = 20.0    # Total day+night length. 20 = vanilla.
seasonLatitude = 48.0      # Central Europe latitude. -90 (short days) to 90 (long days). Serene Seasons detects this automatically.
```

Dynamic Crosshair (configurable in-game via Mod List → Dynamic Crosshair → Config, or `config/dynamiccrosshair.json`):

Dynamic Crosshair uses **YACL** (already in the pack) for its in-game config screen. Recommended settings:

Setting	Value	Effect
<code>hideWhenNotLookingAtAnything</code>	<code>true</code>	Hides crosshair when pointing at sky/air — clean UI during building/exploration
<code>showWhenLookingAtBlock</code>	<code>true</code>	Reappears when looking at a block — signals interactability
<code>showWhenLookingAtEntity</code>	<code>true</code>	Reappears when looking at mobs/players — signals combat target
<code>showWhenHoldingTool</code>	<code>true</code>	Shows crosshair when mining with pickaxe/axe/shovel — shows exact breaking target
<code>showWhenHoldingWeapon</code>	<code>true</code>	Shows crosshair in combat with any weapon

Setting	Value	Effect
<code>crosshairStyle</code>	"CONTEXTUAL"	Changes crosshair shape per context (default dot, block outline for mining, reticle for combat)
<code>crosshairColor</code>	"ENTITY_BASED"	White for neutral/block, red for hostile entities, green for interactable entities

No keybind changes needed — the mod works fully through context detection. YACL provides a searchable settings screen.

All the visual polish, UI improvements, inventory tools, storage, travel, trading, and quality-of-life features that make the game feel great. Install alongside Wave 0.

Visual & Client QoL — What to Expect

These mods change how the game looks and feels. All are client-side and work immediately.

LambDynamicLights

Hold a torch, glowstone, or lantern → it emits light around you. Works in off-hand. **Pro tip:** Hold torch in off-hand while mining. No more placing torches every 5 blocks. Toggle with `Ctrl+L`.

BetterF3

Press `F3` for a cleaner debug HUD. Customize in Mod Menu → BetterF3 → Config. Toggle FPS, coords, biome, light level without clutter.

Sound Physics Remastered

Sound behaves realistically — muffled through walls, echo in caves, directional audio. Works automatically.

AmbientSounds

Biome-aware ambient audio — birds in forests, wind in mountains, water near rivers. Automatic, no config.

The Immersive Music Mod

147 new biome/structure-aware songs with natural silence gaps (unlike Constant Music which forces a 2-minute fade). Plays like vanilla but with much more variety.

Immersive UI

Smooth hotbar sliding, item pickup particles, animated tooltips, menu transitions. Pure polish.

Status Effect Bars Reforged

Small progress bars on status effect icons showing remaining duration. Config via Cloth Config. Auto-hides for beacon effects.

Traveler's Titles

RPG-style title cards when entering biomes, dimensions, or deep caves/high peaks. No config needed.

Better Modlist

Enhanced mod list screen with badges (library/core/addon), categories, and search.

Better Third Person

Decouples camera from character in third-person. Look around independently while walking any direction. Essential for combat awareness and scenic building. **How:** Bind a toggle key in Controls → Better Third Person. Once active, character moves independently of camera.

Tooltip Overhaul

Rounded corners, sharper text, scrollable tooltips, 3D item previews on hover. No config.

First Person Model + 3D Skin Layers

Shows your full body in first-person view, with 3D depth on skin layers (jackets, hats). Requires Not Enough Animations (auto-installed dependency).

Presence Footsteps

Block-aware footstep sounds for every surface — wood creaks, gravel crunches, stone clacks. Works with Sound Physics for realistic acoustics.

Particular Reforged

Ambient visual effects: 3D water splashes, waterfall cascades, fireflies at dusk, falling leaves, cave dust, underwater bubbles, rain ripples. Toggle individual effects in `config/particular.json` if any feel excessive.

Wakes Reforged

Realistic water wakes and splashes when traveling on water. Works with Small Ships (Wave 2) and vanilla boats.

Sounds

170+ new sound effects for UI clicks, item drops, eating, achievements, block actions. Every vanilla action feels tactile. Customize individually in Mod List → Sounds → Config or `config/sounds/`. Conflicts with nothing — complements Sound Physics (environment) perfectly.

Better Days

Controls day/night cycle length. Default is 20-minute day / 10-minute night (2× day length). Config in `config/betterdays-common.toml`:

- `speedMethod = "MINUTES"`
- `daySpeedMinutes = 20, nightSpeedMinutes = 10`
- `enableSleepFeature = true`
- All time effects OFF (`potionEffect = "NEVER", hungerEffect = "NEVER", blockEntityEffect = "NEVER", cropEffect = "NEVER"` — conflicts with Serene Seasons)

SEASON mode: Since this pack includes **Serene Seasons**, you can try `speedMethod = "SEASON"` for immersive day-length variation by season. Experimental — test before committing.

Better Safe Bed

Mobs that can't reach you no longer block sleep. Chill-rule essential.

Beautiful Enchanted Books [Mod Edition]

Unique textures per enchantment type — Protection books look different from Sharpness books. Standalone mod, no resource pack needed.

Overflowing Bars

Health, armor, and toughness bars expand beyond vanilla limits. Essential once you stack Apotheosis bonuses and skill tree HP.

Cherished Worlds

Favorite/pin worlds to the top of your list, prevent accidental deletion. Right-click a world in the list to favorite it.

Day Counter

Day counter title overlay at dawn with milestone celebrations (Day 10, 50, 100...). Appears automatically — no config.

Eating Animation

First-person eating/drinking animation — food moves to your mouth and shrinks. Works with Farmer's Delight foods.

Camera Overhaul

Camera tilting on movement, screen shakes on explosions, mouselook roll, idle sway. Satisfying feedback. Configurable in `config/cameraoverhaul-client.toml`.

[DLS] Dark Loading Screen

Replaces the bright-red loading screen with a dark version. No config.

Inventory & UI — Your Daily Tools

JEI (Just Enough Items)

Your primary learning tool. **Hover any item, press R** to see how to craft it. **Press U** to see what it's used in. Every mod in future waves is learned through JEI first.

Jade + Jade Addons

Look at any block → HUD shows block name, mod source, machine status, crop growth, chest contents. **Jade Addons** extends this with panels for Create (SU/RPM), Mekanism (energy/gas), AE2 (channel info).

Pro tip: Keep Jade enabled. Fastest way to check if a machine is running or a crop is ready.

Conflict fix: Main Menu → Mods → Jade → Config → Mod Settings → Display Bosses → OFF (avoids conflict with Enhanced Boss Bars).

JourneyMap + JourneyMap Integration

- **Minimap** — top-right corner, auto-maps, shows entity dots
- **Fullscreen map** (`J`) — zoomable world map with biome colors, terrain height, waypoints
- **Waypoints** — `B` at your location, or set on the map. Custom icons, colors, groups
- **Entity radar** — shows nearby mobs/animals on minimap
- **Integration** — your placed Waystones appear as icons on JourneyMap automatically

Pro tip: `J` → right-click to set temporary waypoint. Double-click a waypoint name to edit it.

AppleSkin

Hover over food → see exact hunger and saturation values. Pre-eat preview shows dashed outlines on your hunger bar. Saturation overlay shows the hidden second layer. No config.

Mouse Tweaks

Inventory shortcuts: left-click drag → distribute evenly across slots. Right-click drag → place one per slot. Scroll wheel on stack → move one item. No config.

Inventory Essentials

Works alongside Mouse Tweaks:

- `Ctrl+click` a stack → moves one item
- `Shift+Ctrl+click` → moves all items of that type
- Sort buttons on chest/inventory GUIs

Shulker Box Tooltip

Hover over a shulker box in inventory → preview its contents. No config.

Item Borders

Colored rarity borders on items. Common = gray, Uncommon = green, Rare = blue, Epic = purple, Mythic = gold. Helps spot good drops instantly.

Loot Journal

Animated pickup notifications near your hotbar. Session summary on disconnect shows what you gained/lost. Config in Mod Menu → Loot Journal:

- `trackItemPickups = true`
- `trackXpPickups = true`
- `showSummaryOnDisconnect = true`

Reliable Advancements

Overhauled advancements UI — larger screen, pan/zoom, tab sorting, search. Open with `L`. Pairs with Fancy Toasts (animated popups).

Fancy Toasts

Beautiful animated advancement popups. No config.

Polymorph

When two mods have conflicting recipes for the same item, a small button appears next to the crafting output. Click to cycle recipes. Persists per ingredient combination.

Storage & Travel

Sophisticated Storage + Backpacks

- **Barrel** — craft: 1 chest + 8 planks. Upgrade with iron/gold/diamond/netherite upgrades for more slots, pickup, magnet, filter, composting
- **Backpack** — equip in chest armor slot, press `B` to open. Upgradable like barrels
- **Create Integration** (auto-installed) — Sophisticated barrels and backpacks work fully on Create contraptions: upgrades, automation, inventory settings all function while moving

Waystones

Craft a waystone → right-click to claim. Use a Warp Stone (ender pearl + amethyst) to teleport from anywhere. **No XP costs** (`enableCosts = false` in config). Set keybind to `Ctrl+Shift+W`.

GraveStone Mod

Die → grave spawns with all your items. Right-click the grave to retrieve. No item loss, no despawning, 24-hour protection.

Village & Trading

Dynamic Villager Trades

Villager trades change dynamically — villagers specialize in what you buy from them. No more massive trading halls.

Config (`config/dynamicvillagertraders.properties`):

- `global_randomness = 0.7` — faster specialization
- `refresh_delay = 1` — trades change every other restock
- `no_book_duplicates = true` — no duplicate enchantment books

Improved Village Placement

Forces villages to spawn on flat terrain. No more cliff-side or underwater villages. No config.

Session Plan — Wave 0.5

Session	Goal
Session 1 — Setup	Launch, set keybinds, configure Distant Horizons (LOD 128-256, Medium quality). Install Complementary Unbound shader. Enable Alacrity + Fresh Animations texture packs. Walk around spawn to generate LOD data
Session 2 — Survival basics	Punch trees, build a starter shack, find food, get iron. Place your first Waystone. Craft a Sophisticated Storage barrel. Craft a backpack
Session 3 — Comfort	Explore nearby terrain. Find a village. Set up a bed. Light up your base perimeter. Start noticing Traveler's Titles when entering new biomes

Wave 1 — Tech (Create + Mekanism + AE2)

The classic tech trio. Create handles mechanical automation, Mekanism handles industrial processing, AE2 handles digital storage and autocrafting.

Core Tech

Mod	Role
Create	Mechanical automation — rotational power, water wheels, windmills, gear trains, conveyor belts, deployers, sequenced crafters, trains
Mekanism	Industrial processing — ore multiplication (2x→5x), digital miner, superdense energy storage, MekaSuit (flight, invulnerability)
Mekanism Generators	Power generation — heat, wind, solar, gas-burning, bio generators, fission reactor, fusion reactor, turbine
AE2 19.2.17	Digital storage — ME drives, autocrafting, P2P tunnels, spatial storage. The storage backbone
Applied Mekanistics	AE2-Mekanism integration — autocrafting for chemical/gas/fission/fusion recipes, ME chemical storage, P2P chemical tunnels

Note: Mekanism v10 for 1.21.1 NeoForge is split into separate modules. **Mekanism** (core) provides machines, processing, cables, and armor. **Mekanism Generators** provides all power generation. Both must be installed.

Create Addons

Mod	Role
Create Crafts & Additions 1.6.0	Electric motors, alternators, rolling mill, wires/rods, silver/electrum — bridges kinetic SU and Forge Energy (FE)
Create: Power Grid 0.5.5.1	Physics-based electricity simulation — Ohm's law, AC/DC, generators, motors, cables, FE conversion. Separates Create power from SU
Steam 'n' Rails NeoForge 0.2.1	Expanded train system — new tracks (spruce, monorail), semaphores, conductor mob, coupling/decoupling blocks
Create: The Factory Must Grow (TFMG)	Heavy engineering & oil — crude oil drilling, distillation, diesel/gasoline/LPG engines, steel, aluminum, electricity with voltage
Mekanism TFMG Compatibility	85+ recipe bridges — deduplicates lead/steel/sulfur, integrates aluminum/plastic into Mekanism chains
TFMG Energy Converter 26.03	Bidirectional TFMG electricity ↔ FE/AE2 bridge — lets TFMG diesel engines/generators power non-TFMG machines (Mekanism, AE2). Requires Kotlin for Forge
Create: Gunsmithing	Steampunk firearms integrated with Create — craftable using Create machines. Animated guns, attachments, ammo types. Optional Better Combat compatibility

Mod	Role
Create Ornithopter Glider	Mechanical flapping glider with boost-assisted flight — integrates with Create rotational power
Create: Train Utilities 3.0.3	Sliding doors with realistic sounds, platform blocks, station building blocks
Create Railways Navigator	Display boards, route navigation, new schedule entries — search routes and manage train networks
Create: Interiors 0.6.1	Colored chairs with armrest toggles, tray tables, kelp seats — functional furniture for train cars
Create Train Parts 0.4.1	Crossing gates, sliding windows, train steps, slides — decorative and functional blocks for trains and stations
Create: Threaded Trains	Performance — runs train pathfinding on a separate thread, prevents tick lag on complex rail networks

Mekanism Addons

Mod	Role
Advanced Finders	Ore detection devices — directional indicators for nearby ore veins, synchronized multiplayer support
Advanced Chimneys	Smoke ventilation systems — redirect factory emissions outdoors, customizable appearance and behavior

Dependencies: Both Advanced Finders and Advanced Chimneys require [ForgeEndertech](#) library.

How They Work Together

Layer	Mod	
Power generation	Create (water wheels, windmills, steam) + Mekanism (advanced generators, fusion reactor)	
Ore processing	Mekanism — up to 5x ore multiplication with chemical processing	
Factory automation	Create — conveyor belts, deployers, sequenced crafters	
Storage	AE2 — ME drives, storage busses, crafting CPUs	
Transport	Create trains + AE2 P2P tunnels	
Endgame	Mekanism fusion reactor + MekaSuit + AE2 autocrafting	### Ores

Mod	Role
Create (built-in)	Zinc, brass, copper
Mekanism (built-in)	Osmium, tin, lead, uranium, fluorite

Configuration — Wave 1

Mekanism (`config/Mekanism/general.toml`):

- `allowChunkloading = false` — Disable Mekanism's built-in chunk loading. Prevents conflict with the dedicated Chunk Loaders mod and keeps chunk management unified.

Ore generation: All default rates. Osmium at iron-like frequency, tin/lead moderate, uranium/fluorite rarer. Default distribution aligns with the pack's progression curve.

Interdependency notes for Wave 1:

- Mekanism's universal cables can power AE2 controllers via the Energy Acceptor. No adapter mod needed.
- **Create Crafts & Additions** electric motors and alternators create a direct kinetic↔FE bridge, replacing the Create Alternator workaround. Silver and electrum add new material tiers that integrate with Mekanism's ore processing line.
- **Steam 'n' Rails** expands Create's train system with new tracks, semaphores, conductor mobs, and coupling blocks. Essential for moving resources between MineColonies, Mekanism factories, and AE2 storage across a sprawling base.
- **Create train ecosystem:** SnR provides track types (monorail, wooden, wide gauge), semaphores, and the conductor mob. **Train Utilities** adds station architecture (sliding doors, platform blocks). **Railways Navigator** adds display boards and route management. **Interiors** makes train cars usable with seats and tray tables. **Train Parts** adds crossing gates, windows, and steps for detailing. **Threaded Trains** keeps the whole network performant by running pathfinding on a separate thread. Together they cover tracks → stations → navigation → interiors → detailing → performance without overlap.
- Mekanism's 5x ore processing (Chemical Dissolution Chamber → Washer → Crystallizer) outclasses Create's Crushing Wheels (1.5x) and Millstone (1x) for ore multiplication. This is intentional — Create handles mechanical automation and component crafting, Mekanism handles industrial processing. Neither obsoletes the other.
- AE2's Storage Bus on a Sophisticated Storage barrel makes early-mid game storage migration seamless. Place a Storage Bus on your barrel wall, configure it to show contents in the ME terminal, and transition gradually to full digital storage.
- **Sophisticated Storage Create Integration** and **Sophisticated Backpacks Create Integration** (both by P3pp3rF1y) enable full backpack and barrel functionality on Create contraptions — upgrades, automation features (pickup, magnet, feeding), and inventory settings all work while the contraption is in motion. Without these addons, Sophisticated inventories on contraptions are limited to basic item transport.
- **Applied Mekanistics** bridges AE2 autocrafting with Mekanism's chemical processing — ME Chemical Output Buses, ME Chemical Import Buses, and P2P tunnels for chemicals/gases enable autocrafting of Mekanism recipes (including fission/fusion products) directly through the AE2 crafting CPU. Without this mod, Mekanism chemical steps are manual bottlenecks.
- **Create: The Factory Must Grow** adds oil drilling, distillation towers, blast furnaces, and diesel engines — expanding Create's mechanical toolkit into heavy industry. Its steel replaces Mekanism's steel (TFMG's blast furnace is the intended steel source; Mekanism's steel recipe is disabled by the Mekanism TFMG Compat mod). TFMG aluminum and plastic feed into Mekanism's advanced processing chains.
- **Mekanism TFMG Compatibility** resolves 85+ recipe overlaps between TFMG and Mekanism — lead, sulfur, and steel are unified under TFMG's production lines, while Mekanism retains its chemical processing advantage. Both JEI item blacklist config and per-recipe toggles are available for pack tuning.

The core tech trio. Create handles mechanical automation, Mekanism handles industrial processing, AE2 handles digital storage and autocrafting. Install all three core mods plus addons at once.

Core Tech

Create — Mechanical Automation

Create is about **rotational power (SU)**. Machines consume SU to run. If consumption exceeds generation, machines stop.

Learning with Ponder: Hold **W** while hovering over any Create block/item in JEI. An animated tutorial plays showing exactly how it works. **Use Ponder constantly** — it's better than any written guide.

First Create setup:

1. Mine **andesite** — the most important early resource
2. Craft **andesite casing** (andesite + wood nugget in JEI)
3. Craft **cogwheel** and **large cogwheel** — transfer rotational power
4. Place a **hand crank** → attach cogwheel → attach **millstone** → right-click crank to grind wheat/ore manually
5. Build a **water wheel** in flowing water (2–3 block drop) → attach shaft → power 2–3 basic machines (256 SU)

Brass progression (mid-Create): Find zinc ore → mix zinc + copper in a **heated mixer** → brass ingots → brass casing → unlock:

- **Deployer** — places blocks/uses items from inventory. Automates crafting
- **Sequenced Crafter** — multi-step autocrafting sequence
- **Mechanical Arm** — moves items between inventories
- **Chute & Funnel** — vertical item transport with filtering
- **Train Station** — build tracks, assemble trains

Blueprint system: Craft **Blueprint and Quill**. Select an area, save as a blueprint. Place a **Deployer** with the blueprint to auto-build it. Or use the **Schematicannon** — load a blueprint, feed it materials from chests, and it builds the structure for you automatically. This is your “here's a blueprint → build it” tool.

Keybinds:

- Ponder: **W** (hold over block)
- Wrench: right-click
- Goggles: **G** (hold over machine) — shows SU, RPM, stress info
- Toolbox: **T** (hold)

Goggles are essential — craft early, check JEI.

Mekanism — Industrial Processing

Mekanism v10 for 1.21.1 is split into two required modules:

- **Mekanism** (core) — machines, processing, cables, armor
- **Mekanism Generators** — all power generation (heat, wind, solar, gas-burning, bio, fission, fusion)

Finding Osmium: Generates at all underground levels, similar rarity to iron. Mine with stone pickaxe or better.

Tier system: Every machine has 4 tiers. Upgrade by shift-right-clicking with the tier upgrade item.

Tier	Color	Speed	Upgrade
Basic	Orange	1x	Starting
Advanced	Red	2x	Basic + osmium + redstone
Elite	Dark Red	4x	Advanced + diamond + obsidian
Ultimate	Purple	8x	Elite + refined obsidian + alloy

First Mekanism setup:

1. **Heat Generator** — place lava source below it for early power
2. **Metallurgic Infuser** — combine osmium + carbon (coal) → **steel ingots**
3. **Enrichment Chamber** — 1 ore → 2 dust → smelt → 2 ingots (2x processing)

Processing multipliers:

- **2x:** Enrichment Chamber only
- **3x:** Enrichment Chamber → Crusher → Enrichment Chamber again
- **5x:** Full chemical line — Chemical Dissolution Chamber (sulfuric acid) → Washer → Crystallizer → 5 dust

Gas infrastructure (for 5x processing):

- Hydrogen/Oxygen: Electrolytic Separator (split water)
- Sulfuric Acid: Chemical Infuser (sulfur + oxygen)
- Deuterium/Tritium: Fusion reactor fuel (late game)

Gases move through **pressurized tubes**. Use a **Rotary Condensentrator** to convert gas ↔ liquid.

Digital Miner (mid-game): Automated mining machine. Set radius (up to 64 blocks) and filters (whitelist ores). Needs significant power. Tip: set to mine only ores to avoid filling with stone.

Fusion Reactor (endgame): Millions of Joules per tick. Fuel chain: water → Electrolytic Separator → hydrogen → deuterium → tritium → feed both into reactor. Requires significant infrastructure — a Phase 3 project.

Config: `config/Mekanism/general.toml` → `allowChunkloading = false` (delegated to Chunk Loaders mod).

AE2 — Applied Energistics 2

Digital item storage and autocrafting. Everything lives in an electronic network accessed from a single terminal.

Finding Certus Quartz:

- **Meteorites** — circular surface depressions. Dig down to find certus quartz blocks + sky stone chest with free AE2 components
- **Ore veins** — certus quartz ore underground

Processing: certus quartz block → mine → item → craft into dust → smelt → crystal → charge in **AE2 Charger** → charged certus quartz

Channel system: AE2 limits devices per cable. This is the hardest concept for new players.

Cable type	Channels	Notes
ME Glass Cable	8	No indicator
ME Smart Cable	8	Colored lines show usage

Cable type		Channels	Notes	
ME Dense Smart Cable		32	Thicker, for backbone	

- Each ME Controller face provides 32 channels
- Each device (drive, terminal, bus) consumes 1 channel
- Exceed channels → devices show “no power”

Use ME Smart Cable everywhere — colored lines instantly show usage.

First network:

1. **Power:** Build a **vibration chamber** or **energy acceptor**, connect any power source (Mekanism generator works)
2. **Controller:** Craft ME Controller, place it, connect power. One controller = 32 channels
3. **Storage:** Craft **ME Drive + 1k storage cells** (charged certus + redstone + glass), place next to controller
4. **Terminal:** Craft **ME Crafting Terminal** — essential, lets you craft from stored items

Connecting to the world:

- **ME Import Bus** — pulls items from adjacent inventory into network
- **ME Export Bus** — pushes items from network into adjacent inventory
- **Storage Bus** — treats adjacent inventory as part of network (no import/export needed)
- **ME Interface** — exports AND accepts items for autocrafting

Autocrafting:

1. Craft **Patterns** (blank pattern + items in grid)
2. Put patterns in **ME Pattern Provider**
3. Place provider next to **ME Molecular Assembler**
4. Request item from terminal — AE2 pulls ingredients, crafts, returns result

For Mekanism/Create machines: use **ME Interface** instead of pattern provider. Put pattern in interface, it exports ingredients to adjacent machine and accepts the result.

Subnetworks (advanced): When you exceed 32 channels, add a second controller connected via **ME P2P Tunnel** or **ME Quantum Link** for wireless.

Applied Mekanistics

Bridges AE2 autocrafting with Mekanism’s chemical processing. Adds:

- ME Chemical Output Buses / Import Buses
- P2P tunnels for chemicals/gases
- Autocraft Mekanism recipes (including fission/fusion products) through AE2 crafting CPU

Without this mod, Mekanism chemical steps are manual bottlenecks.

Create Addons

Create Crafts & Additions

Electric motors and alternators create a direct **kinetic SU ↔ Forge Energy (FE)** bridge. Also adds:

- **Rolling mill** — processes metal into wires/rods
- **Silver and electrum** — new material tiers that integrate with Mekanism ore processing

Use: Connect an electric motor to your Create network to power it with FE from Mekanism. Or connect an alternator to generate FE from Create rotational power.

Create: Power Grid

Physics-based electricity simulation — Ohm's law, AC/DC, cable loss, transformers. Bridges to FE so Power Grid can power Mekanism/AE2 and vice versa.

Key concepts:

- Generators produce voltage, cables have resistance, machines draw current
- AC travels farther but needs rectification for DC machines
- Use transformers to step voltage up for long runs, then step down locally
- Built-in FE conversion via FE Converter block

When to build: Mid-Phase 2, after you have basic Create power and have outgrown SU. Diesel engines from TFMG feed into Power Grid naturally.

Steam 'n' Rails

Expanded train system:

- New track types: spruce, monorail, wide gauge
- Semaphores for signal control
- Conductor mob (rides trains, interacts with stations)
- Coupling/decoupling blocks

Essential for moving resources between MineColonies, Mekanism factories, and AE2 storage across a sprawling base.

Create: The Factory Must Grow (TFMG)

Heavy engineering and oil processing. Drill for crude oil, refine it, burn fuels in diesel engines.

Oil processing chain:

1. Find oil — surface seepage in desert/plains biomes, or underground pools
2. **Pumpjack** — extracts oil automatically (crafted from brass and steel)
3. **Distillation Tower** (multi-block, 4+ blocks tall) — refines crude oil into diesel, gasoline, LPG, lubricant, asphalt
4. **Cracker** — breaks heavy fractions into lighter fuels
5. **Desulfurizer** — cleans fuel for more efficiency

Diesel engines:

Engine	Fuel	SU Output
Small Diesel	Diesel/gasoline	~2,000 SU
Large Diesel	Diesel/gasoline	~8,000 SU
Gas Turbine	LPG	~16,000 SU

Progression: water wheel → steam engine → diesel engine.

Steel Blast Furnace: Multi-block structure for industrial-scale steel. Faster than Mekanism's Metallurgic Infuser once you have oil power.

Aluminum: Bauxite ore (mid-to-high Y-levels) → multi-block smelter → aluminum ingots. Used in advanced TFMG machines and feeds into Mekanism alloy processing.

TFMG Energy Converter: Bidirectional bridge between TFMG electricity ↔ FE/AE2. Wire diesel generators into your ME system or Mekanism induction matrix. Requires Kotlin for Forge (auto-installed Wave 0 dep).

Mekanism TFMG Compatibility: Resolves 85+ recipe overlaps between TFMG and Mekanism. Steel, lead, sulfur unified under TFMG. Mekanism's steel recipe disabled — use TFMG Blast Furnace instead. Config in [config/mekanismtfgmcompat-common.toml](#).

Create: Gunsmithing

Steampunk firearms crafted using Create machines. Animated guns with attachments and ammo types.

Getting started:

1. Build a **Gunsmithing Table** (check JEI — uses Create brass + wood)
2. Craft **gun frames** — pistol, rifle, shotgun, each with different stats
3. Craft **barrels, grips, magazines** at the mechanical press
4. Assemble at the Gunsmithing Table
5. Craft **ammo** — each gun type uses different ammo (pistol rounds, rifle cartridges, shotgun shells)

Attachments: Scopes, extended magazines, suppressors, bayonets. Attach at the Gunsmithing Table.

Compatibility: Built-in Better Combat support — guns use appropriate animations.

Dependencies: NTGL (gun framework), GeckoLib (3D animation), Framework (utility library) — all auto-installed.

Create Ornithopter Glider

Mechanical flapping glider powered by Create rotational power.

How: Craft the glider (check JEI — brass + leather + feathers). Equip in chest slot. Hold jump to flap (consumes SU from a connected power source or internal buffer). Boost-assisted flight for short bursts.

Use: Early flight before MekaSuit. Pairs with Create windmills or water wheels for power.

Create Train Ecosystem

Six mods work together for a complete train network:

Mod	Role
Steam 'n' Rails	Track types, semaphores, conductor, coupling
Train Utilities	Station architecture — sliding doors, platform blocks
Railways Navigator	Display boards, route management, schedule entries
Interiors	Train car furniture — chairs, tray tables, kelp seats
Train Parts	Crossing gates, sliding windows, train steps, slides
Threaded Trains	Performance — train pathfinding on separate thread

Together: tracks → stations → navigation → interiors → detailing → performance, with no overlap.

Mekanism Addons

Advanced Finders

Ore detection devices with directional indicators for nearby ore veins.

How: Craft a finder (check JEI — tiers for different ores). Hold it — arrows point toward nearest vein, distance shown on HUD. Synchronized in multiplayer.

Advanced Chimneys

Smoke ventilation systems for factories. Redirect emissions outdoors.

How: Craft chimney blocks, place above furnaces/generators. Smoke renders realistically and vents upward. Customizable appearance.

Dependencies: Both require ForgeEndertech library (auto-installed).

How Wave 1 Mods Work Together

Layer	Mod	What's Happening
Power generation	Create + Mekanism Generators	Water wheels → windmills → steam → heat/gas/solar → fission → fusion
Ore processing	Mekanism	2x→3x→5x ore multiplication
Factory automation	Create	Conveyors, deployers, sequenced crafters, mechanical arms
Storage	AE2	ME drives, storage busses, crafting CPUs
Transport	Create trains + AE2 P2P tunnels	Bulk item movement across bases
Endgame	Mekanism fusion + MekaSuit + AE2 autocrafting	God-tier power and automation
Oil/steel	TFMG	Distillation → diesel engines → blast furnace steel
Power bridge	Crafts & Additions + TFMG Energy Converter	Kinetic ↔ FE ↔ TFMG voltage
Rail expansion	Steam 'n' Rails + train ecosystem	Full train network with stations and interiors
Colony defense	Create Big Cannons (Wave 5)	Artillery for base perimeter

Integration highlights:

- Mekanism universal cables power AE2 controllers via Energy Acceptor
 - AE2 Storage Bus on Sophisticated Storage barrel = seamless early-mid storage migration
 - Sophisticated Storage/Backpacks Create Integration = full backpack/barrel functionality on moving contraptions
 - Applied Mekanistics bridges AE2 autocrafting with Mekanism chemical processing
 - TFMG steel replaces Mekanism steel (blast furnace is the intended source)
-

Session Plan — Wave 1

Session	Focus	What to Build
Session 1 — Create basics	Mine andesite. Hand crank + millstone. Water wheel. Grind wheat and ore	256 SU power, basic ore doubling
Session 2 — Mekanism start	Find osmium. Heat Generator + Enrichment Chamber + Metallurgic Infuser. 2x ore processing	Basic Mekanism line, first steel
Session 3 — AE2 discovery	Find certus quartz meteorite. Mine it. Craft dust → crystal → charge. Build AE2 Charger	First certus, charge infrastructure
Session 4 — AE2 network	ME Controller + ME Drive + 1k cells + ME Crafting Terminal. Storage Bus on Sophisticated barrel	First digital storage network
Session 5 — Mid Create	Find zinc. Make brass. Deployer + Sequenced Crafter + Mechanical Arm. Automate a simple recipe	Component automation, brass-tier
Session 6 — Mekanism expand	Add Crusher for 3x ore. Electrolytic Separator for gas. Upgrade to Advanced tier	3x multiplication, gas infrastructure
Session 7 — Integration	Import Bus on Enrichment Chamber → AE2. Export Bus to Create machines. Connect alternator to Mek power	Cross-mod automation bridge
Session 8 — TFMG start	Find oil (desert/plains pools). Pumpjack + Distillation Tower. Small Diesel Engine	Oil refining, massive SU upgrade
Session 9 — TFMG expand	Build Steel Blast Furnace. Mass-produce steel. Find bauxite for aluminum	Industrial steel, aluminum
Session 10+ — Scaling	Digital Miner. Expand AE2 drives. Add Create train between base and outpost. Work toward 5x processing	Automated mining, bulk transport

Pacing tip: You don't need to finish Wave 1 before starting Wave 1.5 (colony) or Wave 2 (exploration). Dip into exploration between factory sessions. The colony takes real-world days to grow — start it early. TFMG is mid-game — start after you have brass Create and basic Mekanism.

Wave 1.5 — Colony Management

MineColonies provides autonomous NPC workers that mine, farm, craft, and guard your base.

Colony Automation

Mod	Role
MineColonies	Autonomous NPC colony — builders, miners, farmers, crafters, guards, research tree
Structurize	Structure building/placement library (MineColonies dependency)
MultiPiston	Multi-block piston library (MineColonies dependency)
BlockUI	UI library (MineColonies dependency)
Domum Ornamentum	Decorative block library (MineColonies dependency)

Interdependency notes for Wave 1.5

- MineColonies guards (deployed via the Town Hall’s “Guard Tower” building) patrol your base perimeter, reducing the need to secure every wall gap manually. Their effectiveness scales with the Guard Tower level and the colony’s skill research. This creates a natural progression — automated base defense grows as your colony does.
- MineColonies Colony Research milestones signal Phase 2 progression readiness. For example, the “Industrial Revolution” research (Colony level 3, 5000 research points) serves as a soft gate — by the time you reach it, you have the colony infrastructure to support Mechanism and AE2. Hard recipe gating (e.g., Atomic Alloy locked behind research) requires a data pack or scripting mod and is not implemented in this pack — the progression pillars in AGENTS.md define the intent, and playtesting determines if soft gating is sufficient.
- Chunk Loaders (Wave 0) are essential for MineColonies — they keep workers active while you explore other dimensions. Place at least one at your colony after setting up your base.

MineColonies provides autonomous NPC workers that mine, farm, craft, and guard your base. Grows alongside your factory.

Getting Started

1. Craft the **Town Hall** and place it on flat ground
2. Right-click → colony GUI opens
3. Hire a **Builder** from the Citizens tab
4. Place a **Builder’s Hut** near the Town Hall
5. Put wood, cobblestone, and tools in the builder’s chest
6. The builder starts constructing from blueprints

Core Worker Huts (build in this order)

Hut	Resource	What it does
Mine	Coal, iron, stone	Digs shaft mine. Passive ore supply
Lumberjack	Wood	Chops trees, replants. Infinite wood
Farmer	Wheat, carrots, potatoes	Plants, grows, harvests. Feeds colony
Guard Tower	Defense	Patrols base perimeter. Scales with colony size
Crafter	Recipes	Crafts items from storage materials

Research Tree

Research		Colony Level	Points	Unlocks	
Basic Industry		2	2000	Better mines, furnace automation	
Advanced Smithing		3	4000	Better worker tools	
Industrial Revolution		3	5000	Atomic Alloy recipe (Mekanism gate)	
Logistics		4	8000	Warehouse, courier routes	

Tech gating: Atomic Alloy (Mekanism ultimate-tier) requires “Industrial Revolution” research. This prevents skipping from Phase 1 straight to Phase 3.

Guard Scaling

- 1 Guard Tower = 20-block radius coverage
- Each additional tower adds overlapping coverage
- Guard equipment upgrades with colony level (leather → iron → diamond)

Integration With Your Tech Base

- **AE2 Storage Bus** on colony warehouse → all colony resources in your ME terminal
- **Create conveyors** feed materials into builder’s chest for automated restocking
- **Farmer’s Delight meals** supply colony kitchen for worker happiness bonuses
- **Chunk Loaders** keep colony running when you’re in other dimensions

Session Plan — Wave 1.5

Session		Focus		What to Do	
Session 1 — Foundation		Town Hall, Builder’s Hut, supply materials		Colony established, first build started	
Session 2 — First workers		Mine hut + Lumberjack hut		Passive coal/iron/wood income	
Session 3 — Farm + Guard		Farmer hut + Guard Tower		Colony self-sufficient, base defended	
Session 4 — Research		Basic Industry research, Crafter hut		First tech unlocked	
Ongoing		Check in every few sessions. Upgrade buildings, expand guards, unlock research		Atomic Alloy ready by Phase 2	

Tip: The colony is a long-term background investment. Start it early, check periodically, let it grow.

Wave 2 — Exploration (World & Dimensions)

Overworld Worldgen

Mod	Role
Terralith	Overhauled Overworld biomes
Tectonic	Better terrain shape — deeper valleys, taller mountains, cliffs

Environment

Mod	Role
Serene Seasons	Seasonal foliage colors, temperature shifts, crop growth cycles

Serene Seasons adds four seasons (Spring, Summer, Autumn, Winter) with visual foliage changes, temperature shifts, and seasonal crop growth modifiers. Crop impact is enabled — planting in the wrong season reduces yield, encouraging greenhouse planning. Temperature damage is **disabled** — no freezing or heat damage, preserving the chill rule.

Underground & Ocean

Mod	Role
TerraBlender (NeoForge) — REQUIRED	Worldgen biome library — required by Darker Depths 2.1+ for biome placement
Darker Depths	Underground expansion — 3 new cave biomes, mobs, Forsaken Bronze, Living Crystals
Upgrade Aquatic	Ocean expansion — Thrasher mob, Ocean Ravines, Prismarine Coral, renewable sand

Interdependency note — Darker Depths + TerraBlender: Darker Depths 2.1+ requires a worldgen API to register its cave biomes. Either **TerraBlender** (NeoForge) or **Biolith** must be installed. TerraBlender NeoForge is the recommended choice (available for 1.21.1 on both CurseForge and Modrinth). Without it, Darker Depths crashes on startup with [Missing Worldgen API! Install TerraBlender or Biolith](#).

Interdependency note — Darker Depths + YUNG's Better Caves: Darker Depths adds its own cave biomes (Molten Cavern, Sandy Catacombs, Crystal Cavern) that generate alongside YUNG's Better Caves' enlarged cave systems. Darker Depths biomes replace vanilla cave sections with themed variants, while YUNG's Better Caves changes the shape and connectivity of caves — they layer without direct conflict.

Interdependency note — Upgrade Aquatic + YUNG's Better Ocean Monuments: Upgrade Aquatic adds ocean mobs, coral variants, and ocean-floor features. It does not modify Ocean Monuments — YUNG's Better Ocean Monuments replaces monument structure generation without conflict. Upgrade Aquatic items (thrasher teeth, prismarine coral) integrate naturally with Ocean's Delight (Wave 4) for seafood recipes.

Structures

Mod	Role
YUNG's Better Dungeons	Overhauled dungeons

Mod	Role
YUNG's Better Mineshafts	Improved mineshafts
YUNG's Better Strongholds	Larger, more complex strongholds
YUNG's Better Desert Temples	Multi-room desert temples
YUNG's Better Jungle Temples	Jungle temples with traps
YUNG's Better Ocean Monuments	Redesigned ocean monuments
YUNG's Better Witch Huts	Expanded witch huts
YUNG's Better Nether Fortresses	Improved nether fortresses
YUNG's Better Caves	Larger, more varied caves
Structory	New hand-crafted structures
Towns & Towers	Better village generation

Dimensions

Mod	Role
The Aether	Sky dimension — safe islands, gear progression, dungeons with bosses
Deeper and Darker	The “Otherside” dimension — deep dark themed, warden gear
Twilight Forest	Classic dimension — 6+ bosses with progression gates, unique loot, enchanted forest biomes

End Overhaul

Mod	Role
BetterEnd NeoForge v21.0.25	Full End overhaul — 24+ biomes, new mobs, gear, custom music
Moog's End Structures v2.0.3	New structures to fill the End dimension
Nullscape (data pack)	End terrain overhaul — surreal alien landscapes, new sky colors

Nullscape: A data pack that transforms the End’s terrain into alien shapes (floating islands, spires, arches, voids) with unique sky colors per biome. **Compatible with BetterEnd** — Nullscape’s own documentation states it works with BetterEnd specifically (it is incompatible with most End worldgen mods, but BetterEnd is the documented exception). Nullscape handles terrain shape and sky; BetterEnd handles biomes, mobs, gear, and structures. Install as a data pack in the world’s [datapacks/](#) folder.

Navigation

Mod	Role
Explorer's Compass	Craftable structure/biome finder

Mobility

Mod	Role
Immersive Aircraft 1.1.5	Flyable aircraft — biplane, airship, cargo airship, warship, gyrocopter. Upgradable, colorable

Mod	Role
Small Ships 2.0.0-b2.1	Sailable ships — brigantine, cog, drakkar, galley. Build, crew, and navigate; moveable bases on water. Beta (NeoForge)

Configuration — Wave 2

Aether (`aether-common.toml`):

- `"Use default Curios' menu" = true` — Uses Curios API menu instead of Aether's separate accessories screen.
- `"Gives player Aether Portal Frame item" = false` — Player must build the portal manually.
- `"Gives starting loot on entry" = true` — Grants Book of Lore and Golden Parachutes on first entry.
- `"Show Patreon message" = false` — Removes Patreon nag screen.
- `"Add Temporary Freezing automatically" = false` — Disables Aether's temporary freezing datapack.
- `"Add Ruined Portals automatically" = false` — Disables ruined portal datapack.

Interdependency note — Immersive Aircraft + Aether: Aircraft can fly in the Aether dimension. An airship is a natural way to navigate Aether islands — it pairs well with the dimension's sky-island geography.

Interdependency note — Twilight Forest + Apotheosis: Apotheosis boss affixes (Wave 3) apply to Twilight Forest bosses. A Naga with the "Tough" affix is significantly harder. This is intended — it keeps dimension bosses relevant even with late-game gear. If a boss feels too strong, the player can return after better equipment or use Apotheosis gems/runes for the edge.

Interdependency note — Towns & Towers + Cristel Lib: Towns & Towers requires **Cristel Lib** (included in Wave 2 Dependencies above). Without it the game will crash at startup. Cristel Lib provides structure config and runtime datapack support that Towns & Towers relies on for its biome-specific village variants.

The world is dramatically transformed. Terralith overhauls biomes, Tectonic reshapes terrain, YUNG's expands every structure, and three new dimensions open up.

Overworld

Terrain — Terralith + Tectonic

Aspect	What changed	What to look for
Mountains	Taller, steeper. Painted mountains (orange/white layers), volcanic peaks	Build in mountain valleys
Oceans	Deeper, trench-like drops offshore	Use Small Ships (below) to cross
Biomes	80+ overhauled — redwood forests (200+ block trees), cherry groves, floating islands	Cherry groves for vibes, redwoods for wood
Caves	YUNG's Better Caves — wider tunnels, deeper chasms	Easier to navigate, more impressive
Rivers	Deeper valleys from Tectonic, shattered savannahs	Follow rivers for scenic travel

Pro tip: These worldgen mods work best in newly generated chunks. Every direction has something different.

Environment — Serene Seasons

Four seasons with visual foliage changes and crop growth modifiers:

- **Spring/Summer/Autumn/Winter** — foliage colors shift realistically
- **Crop impact ON** — planting in wrong season reduces yield. Plan greenhouses for off-season crops
- **Temperature damage OFF** — no freezing or heat damage (chill rule preserved)

Integration: Farmer's Delight crops are affected by seasons. Build glass greenhouses or use Season Sensor blocks to track optimal planting times.

YUNG's Overhauled Structures

Every vanilla structure is expanded:

Structure	What's new	Loot
Better Dungeons	Multi-room with spawners, traps, treasure	Early iron, gold, enchanted books
Better Mineshafts	Multi-level with support beams, loot caches	Rails, minecarts, ore samples
Better Strongholds	Deeper portal room, more libraries/prisons	Enchanted books, mid-game exploration
Better Desert Temples	Multi-room underground, traps, vault	High-value vault loot
Better Jungle Temples	Vertical temple, redstone puzzles	Puzzle-solving loot
Better Ocean Monuments	New rooms, more elder guardians	Better sponge room loot
Better Witch Huts	Expanded hut with potion lab	Brewing supplies, potions
Better Nether Fortresses	Districts (blaze, wither skeleton, strider)	Nether wart, blaze rods, skulls

Structory + Towns & Towers

- **Structory** — dozens of hand-crafted structures: camps, shrines, castles, towers, walled towns
- **Towns & Towers** — villages generate with unique architecture per biome style. Each biome gets its own building designs

What to do: Explore! Mark interesting structures on your JourneyMap (fullscreen map → right-click to set waypoint). Return with an Explorer's Compass.

Explorer's Compass — Navigation Tool

1. Craft: 1 compass + 4 iron ingots in a + pattern
2. Right-click → GUI opens
3. Search tab: type biome or structure name (partial matching — “cherry” finds all cherry biomes)
4. Select an entry → compass points to nearest instance, distance shown
5. Follow the arrow in real-time
6. When found, add a JourneyMap waypoint

Underground & Ocean

Darker Depths — Expanded Underground

3 unique cave biomes deep underground:

Biome	Depth	What to expect	Key resources
Molten Cavern	Deep (Y < 0)	Lava pools, geysers, amber clusters	Living Crystals (feed diamond to grow), Crystal Melons
Sandy Catacombs	Mid (Y 0-20)	Petrified roots, aridrock, Void Soul mobs	Void Souls (bottle them), Forsaken Bronze
Crystal Cavern	Any depth	Growing crystals, glowing ambience	Crystal materials for decorative/utility

Living Crystals: Found dead in Molten Cavern. Feed a diamond → reawakens, grows, spreads to nearby diamond ore. Crystal Melons (off-hand) = unbreakable tools with +20% speed/damage for 5 minutes.

Void Souls: Kill a Void Soul mob → becomes invulnerable orb (60 seconds). Capture in glass bottle. Use amber to prevent expiration. Forsaken Bronze from void-touched materials has death-related abilities.

Compatibility: Darker Depths biomes generate inside YUNG's Better Caves systems — they layer without conflict.

Required dependency: TerraBlender NeoForge (auto-installed). Darker Depths 2.1+ crashes without it.

Upgrade Aquatic — Richer Oceans

Feature	What it does
Thrasher	Fantasy shark-like mob. Uses sonar, thrashes prey. Drops thrasher teeth for trident crafting
Ocean Ravines	Underwater ravines with Prismarine Coral — harvest for decorative blocks
Nautilus	Passive mob. Drops shells for conduit crafting
Renewable sand	Bubble columns below sandstone convert stone to sand

Integration: Thrasher teeth and prismarine coral integrate with Ocean's Delight (Wave 4) for seafood recipes. Compatible with YUNG's Better Ocean Monuments.

Mobility

Immersive Aircraft

Vanilla-style flyable aircraft. Mid-game (requires iron + basic Create/Mek materials).

Vehicle	Fuel	Seats	Cargo	Best for
Biplane	Coal/charcoal	1	None	Fast personal flight, combat
Airship	Coal/charcoal	3	9 slots	Exploration with friends
Cargo Airship	Coal/charcoal	1	27 slots	Material transport

Vehicle	Fuel	Seats	Cargo	Best for
Warship	Coal/charcoal	3	9 slots	Armed combat, defense
Gyrocopter	Coal/charcoal	1	None	Agile short-range flight

How: Craft at table (check JEI — iron + wool/leather), place on ground, right-click to enter. [W](#) to ascend, [S](#) to descend, [A/D](#) to rotate. Any furnace fuel works.

Integration: Aircraft work in the **Aether** — fly between sky islands without bridges. Park indoors or on high platforms (mobs damage parked aircraft).

Small Ships

Buildable, sailable ships. **Phase 1 accessible** — requires wood, wool, iron.

Ship	Crew	Cargo	Best for
Brigantine	1	18 slots	All-purpose sailing
Cog	2	27 slots	Cargo transport
Drakkar	2	9 slots	Fast travel, combat
Galley	1	9 slots	Early personal vessel

How: Craft hull (wood) + mast (wood + wool) + helm. Right-click water to place. Right-click helm to board. [W/S](#) to move, [A/D](#) to steer. Speed depends on wind direction. Dismount to park — ships float where left. Right-click with lead to tow.

Note: Beta for NeoForge (2.0.0-b2.1). Core features work; minor bugs expected.

Dimensions

The Aether — Sky Dimension

Safe, beautiful sky islands. Gentle adventure curve.

Portal: 4×5 glowstone frame (like Nether portal) → fill with water bucket → step through.

Progression:

- **Phase 1:** Spawn on sky island. Collect **ambrosium** (Aether coal) and **zanite** (Aether iron — mining level increases as durability drops)
- **Phase 2:** Bronze dungeon → defeat **Slider** (hit when blue, avoid when red) → Golden Amber
- **Phase 3:** Silver dungeon → defeat **Valkyrie Queen** → Valkyrie tools (flight-enabled weapons), **Gravitite** (floats upward when right-clicked)
- **Phase 4:** Highest crystal islands → **Sun Spirit** — bring fire resistance

Tip: Not required for any tech progression. Visit when you want a change of pace. MekaSuit trivializes all Aether combat.

Config (`aether-common.toml`):

- `"Use default Curios' menu" = true` — uses Curios API menu
- `"Gives player Aether Portal Frame item" = false` — build portal manually
- `"Gives starting loot on entry" = true` — Book of Lore + Golden Parachutes on first entry
- `"Show Patreon message" = false`
- `"Add Temporary Freezing automatically" = false`
- `"Add Ruined Portals automatically" = false`

Twilight Forest — Progression Dimension

Dark, enchanted dimension with strict boss progression. Each boss kill unlocks the next area.

Portal: Find a 2×2 water pool → surround with 12 flowers (any type) → throw a **diamond** into the pool → lightning strikes → step into the water.

Note: One-way down. To return, craft a **Twilight Portal** block or use a Waystone (place one immediately on arrival).

Boss progression:

Order	Boss	Location	Strategy	Unlock
1	Naga	Naga Courtyard	Strafe charges, hit tail segment	Lich Tower
2	Lich	Lich Tower	Dodge projectiles, hit during cooldown. Shield reflects projectiles	Twilight Swamp
3	Minoshroom	Labyrinth	Find minotaur room, kill it, grab Meef Stroganoff	Firefly Forest
4	Hydra	Hydra Lair	Dodge fireballs, melee heads. Drops Fiery Tears	Ur-Ghast fight
5	Ur-Ghast	Dark Tower	Shoot tentacles, then main body. Dark Forest has blindness — bring milk	Snowy Forest + Highlands
6	Snow Queen	Aurora Palace	Push into pillars to break shield. Avoid ice arrows	Final Castle
7	Final Castle	Highlands	Multi-boss fight. Needs weapons from all prior bosses	Giant's Pick (5×3 mining area)

Key items:

- **Ironwood** gear — smelt ironwood ingots from iron + magic beans. Good starter armor
- **Fiery** gear — iron + blaze powder. Mid-tier weapons
- **Knightmetal** — from Lich Tower
- **Phantom armor** — creative flight for short bursts. From Ur-Ghast tentacles
- **Giant's Pick** — 5×3×5 mining area. From Final Castle

Tips:

- Always twilight — no day/night cycle. Mobs spawn any time
- Place **Waystones** frequently
- **Twilight Forest Maze Map** (paper + torch) reveals maze/castle layouts
- Bring **milk buckets** for Dark Forest blindness
- Your AE2 wireless terminal, Mekanism gear, and Create tools all work here

Integration: Apotheosis boss affixes apply to Twilight Forest bosses. A Naga with “Tough” affix is significantly harder — intended to keep dimension bosses relevant even with late-game gear.

Deeper and Darker — The Otherside

Deep-dark-themed dimension accessed through the Deep Dark.

How to enter:

1. Find a **Deep Dark** biome (Y=-40 or below)
2. Locate an **Ancient City** (Explorer's Compass → search "ancient city")
3. Kill the **Warden** — drops a **Warden Heart**
4. Craft: Warden Heart + 8 sculk → **Otherside Portal Catalyst**
5. Build a 4×5 frame of **reinforced deepslate** → right-click catalyst on frame → step through

Warden strategy: Detects by sound. Use wool to silence footsteps. Sneak near sculk shriekers. Throw snowballs to distract. 500 HP, slow attacks — hit and run.

What to expect:

- 4 new biomes — sculk forests, frozen deep, expanded ancient cities
- 8 new mobs — sculk variants, stalker (ranged), shriek worm (ambush), **Sculk Leech** (heals on damage)
- **Miniboss:** Shattered Warden — drops **Sculk Transmitter** components
- **Key loot: Sculk Transmitter** — Elytra upgrade for mid-flight teleportation. Combine at smithing table

Gear: Warden-themed armor/tools ≈ diamond tier with unique abilities (sound-based attacks, silence fields).

Difficulty: Harder than Aether, easier than Twilight Forest's Dark Forest. Come with iron-tier armor + decent weapon.

End Overhaul

After defeating the Ender Dragon, the End becomes a full dimension.

BetterEnd NeoForge adds 24+ biomes:

Biome type	Description	Resources
Floating islands	Large islands with unique vegetation, end lakes	New wood types, building blocks
Crystal caves	Underground caverns with glowing crystals	Crystal materials
Mushroom fields	Giant end-themed mushrooms	Magic crafting materials
Ice biomes	Frozen wastelands	—
Chorus forests	Dense chorus plant forests	Chorus flowers, new food

Moog's End Structures adds towers, ruins, and dungeons with loot better than vanilla End cities.

Nullscape (data pack) overhauls End terrain into surreal alien shapes — floating islands, spires, arches, void pits, unique sky colors per biome. **Compatible with BetterEnd** (Nullscape's documented exception). Install in [datapacks/](#) folder.

- Nullscape handles terrain and sky
- BetterEnd handles biomes, mobs, gear, and structures

Ender's Delight (Wave 4) adds food cooked from End ingredients — useful status effects for End exploration.

Progression:

- Post-Dragon: Explore outer islands for new materials and tier-appropriate gear
- Mid-End: Tackle Moog's structures for unique weapons
- Endgame: Build an End base with rarest biomes

Tip: Place a Waystone immediately. Build a small AE2 subnet with wireless terminal to access main base storage.

Session Plan — Wave 2

Session	Focus	What to Do
Session 1 — Scout	Craft Explorer's Compass. Search for cherry grove or redwood forest. Mark waypoints on JourneyMap. Loot a YUNG's dungeon	First taste of transformed world
Session 2 — Aether	Gather glowstone. Build Aether portal. Explore first sky island, collect ambrosium and zanite	Safe new dimension
Session 3 — Twilight Forest	Find 2×2 pool, surround with flowers, throw diamond. Enter, place Waystone. Defeat the Naga	First Twilight boss
Session 4+ — Deep exploration	Work through Twilight Forest bosses. Visit Deeper and Darker (find Ancient City). Explore End post-Dragon	Boss loot, dimension resources
Session 5 — Aircraft	Craft a biplane or airship (iron + wool). Explore Terralith biomes from the air	Fast travel, scenic views

Pacing tip: Explore between factory builds. Grab your compass and walk in a random direction when waiting for Mekanism to process or Create to assemble.

Wave 2.5 — Building Your Home

You’ve explored the world, found stunning biomes, and gathered materials. Now it’s time to stop surviving and start **living**. This section covers how to build a home you’re proud of — no creative mode required, no artistic talent assumed. Just a few principles that separate “dirt hut” from “place I actually want to come back to.”

Building is a core pillar of this modpack. At any point you can stop optimizing factories and just build something beautiful. This guide works at any stage — but it’s especially satisfying now, when you have enough materials to be selective but haven’t yet automated everything.

Finding Your Build Site

Before placing a single block, find a location that inspires you. You have tools for this:

Tool	How	Why
Explorer’s Compass	Craft: 1 compass + 4 iron (+pattern). Right-click → search “cherry”, “redwood”, “meadow”	Find specific biomes by name
JourneyMap	Press J → explore the fullscreen map	See terrain, biomes, and your marked waypoints
Small Ship	Sail along coastlines	Discover waterfronts you’d miss on foot
Biplane	Fly low over terrain	Spot valley builds, mountain perches, river mouths

What to look for:

- **Waterfront** — river banks, ocean cliffs, lake shores. Water reflects builds beautifully with shaders
- **Mountain valleys** — enclosed, dramatic, defensible. Terralith makes these spectacular
- **Forest clearings** — surrounded by trees feels cozy and sheltered
- **Cliff faces** — cut into the hillside for a build that looks permanent, not placed
- **Cherry groves** — pink petals + mountain backdrop = instant atmosphere

Pro tip: Place a **Waystone** at your chosen site before you start building. You’ll want to teleport back with materials.

The Block Palette Rule

The single biggest upgrade from “meh” to “wow” is using **3–5 complementary blocks** instead of one. Every great build you’ve ever seen uses a palette. Here’s the formula:

Role	Percentage	Purpose
Primary	~50%	Dominant wall material — defines the build
Secondary	~30%	Complementary texture — similar tone, different surface
Accent	~20%	Detail material — corners, trim, highlights, pops of color

Finding Palettes: BlockPalettes.com

BlockPalettes.com — the single best resource for block combination inspiration. Browse hundreds of community-created palettes, each showing exactly which blocks work together. Filter by style, color, or mood.

How to use it:

1. Open blockpalettes.com/palettes
2. Browse featured palettes or filter by color/theme
3. Click a palette to see all blocks in the combination
4. Test it: place a 5×5 section in a flat area, step back, see if it works
5. If any block fights the others, swap it out

Starter Palettes for This Modpack

These palettes use blocks you can gather in Waves 0–2. Each fits a different biome and vibe:

Medieval Stone — stone brick + cracked stone brick + mossy stone brick + cobblestone (corners/base)

Cozy Cottage — oak planks + spruce planks + cobblestone (chimney/base) + oak leaves (roofline trim)

Desert Sandstone — sandstone + smooth sandstone + cut sandstone + terracotta accents

Nordic Cabin — spruce planks + stripped spruce log + stone + dark oak stairs (roof)

Cherry Zen — cherry planks + cherry log + stone brick + pink petals (garden)

Mountain Fortress — stone brick + deepslate tiles + tuff bricks + copper (accents/patina)

Tropical — jungle planks + bamboo mosaic + mangrove propagules + sand

Endstone Palace — purpur block + purpur pillar + end stone bricks + chorus flower (End builds)

Tip: Open BlockPalettes.com in your browser while you build. When you're stuck on what goes together, browse palettes until something clicks.

Depth and Layering

Flat walls are the second most common beginner mistake (after single-block builds). A flat wall looks like a placeholder. A layered wall looks **built**.

Five Techniques That Transform Flat Walls

Technique	How to do it	Why it works
Inset windows	Push glass back one block into the wall instead of flush with the surface	Creates shadow lines that frame windows
Outcropped corners	Extend corner blocks one block forward from the main wall plane	Breaks up 90-degree edges, adds structural feel
Alternating depth	Every 3–4 blocks along a wall, push one column back or pull one forward	Wall reads as textured rather than uniform
Stair/slab detailing	Upside-down stairs under windowsills, stairs as roofline trim, slabs as half-height ledges	Catches light and shadow in ways full blocks can't
Overhanging roofline	Always extend the roof 1–2 blocks past the wall on all sides	Creates shadow line that defines where building ends and sky begins

Fastest single improvement: Add vertical **log pillars** every 3–5 blocks along exterior walls. Logs imply structural support and instantly break up flat surfaces. Works with every style.

Roof Design

A bad roof ruins a good build. A good roof makes even a simple box look intentional.

Three Roof Styles (Pick One)

Style	Shape	Best for	How
A-frame (pitched)	Triangle cross-section	Cottages, cabins, medieval	Stairs ascending from each wall meeting at a ridge. Overhang 1 block on all sides
Flat with parapet	Flat top, raised edges	Modern, desert, fortress	Slabs or full blocks, 1-block wall around perimeter
Stepped pyramid	Layered squares getting smaller	Asian, temples, fantasy	Each layer 2 blocks smaller than the one below

Material rule: Roof material should **contrast** with wall material. Dark roof on light walls, or vice versa. Spruce stairs on oak walls. Deepslate on stone brick. Blackstone on sandstone.

Common mistake: Roof too small. The roof should be wider and taller than feels “necessary.” An oversized roof looks dramatic; an undersized one looks like a hat that doesn’t fit.

Interior Design

An empty interior is as bad as a bad exterior. Minecraft has no furniture mod yet (Wave 6 adds Handcrafted), but vanilla blocks create convincing rooms.

Lighting (Critical)

Never place torches on interior walls. They look improvised. Better options:

Light source	Look	How
Lanterns	Warm, architectural	Hang from ceiling via fence post
Glow berries	Organic, trailing	Place on ceiling, let vines trail down
Sea lanterns	Clean, neutral	Modern or sci-fi builds
Jack o’lanterns	Hidden under carpets	Floor lighting without visible sources
Froglights	Subtle, colorful	Pearlescent/verdant/ochre variants from Aether or Overworld

Functional Rooms

Room	Key blocks	Details
Kitchen	Smoker, campfire, barrels, chests	Barrel = cabinets. Chest with food item frame = pantry. Cooking pot over campfire
Bedroom	Bed, crafting table, bookshelf, lantern	Banner on wall above bed as headboard. Carpet at foot of bed

Room	Key blocks	Details
Library	Bookshelves, lectern, quill	Cover walls with bookshelves for instant warmth
Storage	Sophisticated Storage barrels, chests	Organized rows. Sign labels on each barrel
Workshop	Crafting table, furnace, anvil	Industrial feel — stone walls, iron accents

Pro tip: Place **item frames** on walls with maps, banners, or shields inside. They function as paintings with more variety.

Blending Into the Landscape

A build that looks dropped from the sky is unfinished, no matter how detailed the structure itself is.

Key principles:

- **Use biome-appropriate materials** — spruce in cold biomes, sandstone in desert, dark oak near jungle edges. The build should look like it was built from local materials, not imported
- **Never leave flat grass at the base** — add a slight slope toward the entrance, gravel/cobblestone paths leading away, small dirt mounds or rock outcroppings at corners
- **Plant trees near the build** — a cluster of 2–3 trees on one side breaks the silhouette and gives context. Use the biome's native trees
- **Cut into hillsides** — a build emerging from a hillside looks permanent; one sitting on top looks temporary
- **Paths and surroundings** — gravel paths, stone brick walkways, and lantern posts leading away from the entrance sell the idea that people live here
- **Never run more than 7 blocks in a straight line** when shaping terrain — curves look natural, straight lines look artificial

Modpack-Specific Building Tools

You have tools that vanilla players don't. Use them:

Tool	What it does for building
Waystones	Place one at your build site. Teleport home for materials, teleport back. No more carrying 3 inventories of blocks across 500 blocks
Sophisticated Storage	Barrels near your build site hold materials organized by type. Label with signs
Backpack	Carry your palette blocks in your backpack while building. Press B to swap materials on the go
Create Blueprint	Craft Blueprint and Quill. Select your finished build → save as blueprint → use Schematicannon to auto-rebuild it elsewhere with materials from chests
Building Wands (Wave 6)	Place walls, fill areas, build circles/cylinders in bulk. Wait for Wave 6 for this one
JourneyMap waypoints	Mark your build site, mark material locations, mark inspiration you find while exploring
Dynamic Lights (Ctrl+L)	Hold a lantern while building at night. No more stopping to place torches

Quick-Start: Your First Real House

If you've never built anything beyond a dirt hut, follow these steps for your first proper home:

1. **Pick a palette** — Open [BlockPalettes.com](https://blockpalettes.com), find 4 blocks you like. Test in a 5×5 section
2. **Lay the foundation** — Cobblestone or stone brick ring, 7×9 interior minimum (small but not cramped)
3. **Raise the walls** — Use your primary block. Leave gaps for windows (2-block wide minimum)
4. **Add depth** — Push windows back 1 block. Add log pillars at corners and every 4 blocks
5. **Build the roof** — A-frame with stairs. Overhang 1 block. Use a contrasting material
6. **Add the door** — Center of one wall. Stairs as a small porch step in front
7. **Light it up** — Lanterns from ceiling, not torches on walls
8. **Furnish basics** — Bed, crafting table, furnace, 2–4 barrels for storage
9. **Blend the exterior** — Paths leading away, trees nearby, flowers at the entrance, leaves around the roofline
10. **Step back and admire** — You just built something worth coming home to

Community Resources

Resource	URL	What it's for
BlockPalettes	blockpalettes.com	Block color combinations and palette inspiration
Building Guide App	buildingguide.app	Step-by-step tutorials with materials lists (128+ builds)
r/MinecraftBuilds	reddit.com/r/MinecraftBuilds	Community showcase, inspiration, feedback
r/MinecraftHouses	reddit.com/r/MinecraftHouses	House-specific builds and ideas
r/MinecraftTutorials	reddit.com/r/MinecraftTutorials	How-to guides for specific techniques
Minecraft Wiki — Building	minecraft.fandom.com/wiki/Building	Official guides on terraforming, furniture, layout

Session Plan — Wave 2.5

Session	Focus	What to Build
Session 1 — Site selection	Explore with Explorer's Compass. Find a biome you love. Place Waystone. Mark on JourneyMap	Your future home location
Session 2 — Palette + foundation	Browse BlockPalettes.com. Pick 4 blocks. Lay foundation ring. Raise walls to 4 blocks high	Shell of your first real house
Session 3 — Roof + depth	Build A-frame roof. Add log pillars. Inset windows. Add door	Recognizable house shape

Session		Focus		What to Build	
Session 4 — Interior		Lanterns from ceiling. Bed, crafting table, furnace. Barrels for storage. Item frames on walls		Livable interior	
Session 5 — Landscaping		Paths, trees, flowers, leaf trim around roofline. Step back and see the difference		Home in a world	

Tip: Don't aim for perfection on your first build. Aim for "better than a dirt hut." You can always expand, renovate, or rebuild later. The skills transfer — your second build will be twice as good.

Wave 3 — Equipment Magic & RPG

RPG Progression

Mod	Role
Skill Tree (RPG Series)	XP leveling — spend points on attributes and passives
Pufferfish's Skills	Underlying skill system framework

Equipment

Mod	Role
Simply Swords	60+ unique melee weapons with passive/activated abilities
Too Many Bows	30+ unique bows with special abilities and custom attributes
Spartan Weaponry Unofficial	Spears, halberds, pikes, lances, throwing knives — fills the polearm gap
Relics RPG	Legendary accessories with unique abilities
Runes	Socketable runes for weapons/tools
Curios API	Accessory slots (rings, amulets, belts, gloves)

Loot & Affixes

Mod	Role
Apotheosis	Rarity tiers, random affixes, gem socketing, enchanting overhaul. Replaces Savaru's Affixology with more depth

Configuration — Wave 3

Apotheosis — the most config-heavy major mod. Apotheosis is split into modules, each with its own config:

Adventure module — affixes, gems, boss modifiers, rogue spawners:

- Keep all affix types enabled: Ruthless (crit), Arcane (spell), Sturdy (armor), Nimble (speed), Plentiful (mining), Titanic (HP). Each supports a different playstyle.
- Gem tiers enabled: Rough, Cut, Flawless. The gem progression (Rough → Cut → Flawless) provides a satisfying vertical upgrade path. Each tier unlockable through Apotheosis's reforging mechanic.
- Gem types cover all primary stats: melee damage, projectile damage, magic damage, armor, max HP, mining speed, movement speed, luck. This ensures every build archetype benefits from gem-socketed gear.
- Socket count: Gear drops with 1-2 sockets by default. Reforging at the Smithing Table can add a 3rd socket at high rarity (Mythic+). This prevents socket overload while keeping the system meaningful.
- Boss modifiers: all active. Apothic Invaders and Elites spawn naturally with random affixes. They always drop affix items — this is the primary source of mythic gear.
- Rogue Spawners: enabled. Underground spawner+chest combos. Risk vs reward.
- Affix item probability: default rates (~5% base for rare+, scaling with luck and difficulty). Common-enough to feel rewarding, rare-enough to keep each upgrade exciting.

Enchanting module — [config/apotheosis/enchanting.cfg](#):

- Keep all enchantment overhauls enabled: higher level cap (up to level 150+), tiered bookshelves (Hell, Seashelf, Endshelf), Library of Alexandria, new enchantments (Chromatic, Mounting, Stable Footing, etc.).
- The Library of Alexandria stores all enchanted books in one block — essential for enchanting organization at scale.
- Enchantment max levels increased for many vanilla enchantments: Protection can reach level 8, Sharpness level 6, Fortune level 5, Looting level 5, etc. These higher caps match the power fantasy curve — gear enchanted at max level should feel mythic.

Potion module:

- Potion Charms enabled: craft a charm from blaze powder + potions, equip in offhand for passive effect. Reduces inventory clutter from potion bottles.
- Potion augmentation allowed: glowing, brewing, changing base type into splash/lingering at standard ratios. Augmented potions can have higher level caps (e.g., Potion of Strength III).

Spawner module:

- Spawner capture and modification enabled. Silk-touch a spawner with a pickaxe, then modify it (entity type, spawn delay, player range, min/max entities). Key for XP farms and mob-based automation.
- Spawner stats are fully configurable after placement through the spawner's GUI (right-click with an empty hand). Players can fine-tune delay, count, range, and conditions per spawner — no external tool needed.

Village module:

- Wandering trader improvements enabled. Traders can sell Apotheosis items (gems, affix gear, enchanted books) at elevated prices. Adds a secondary acquisition path for players who prefer trading over combat.

Deadly module: Enabled. Apotheosis's Deadly module adds randomly generated death-traps to the world (sudden spikes, poison darts, instakill rooms, lava traps). This adds risk-reward tension to structure exploration — the deadliest rooms hold the best loot. The Deadly module's traps are telegraphed and avoidable with caution, and GraveStone Mod keeps your items safe if you do trigger one, so the chill rule is preserved.

Reforging costs — Apotheosis's reforging mechanic (reroll affixes with gem dust) uses default material costs. As the player progresses, reforging costs scale:

- Common→Uncommon: cheap (a few gem dust)
- Rare→Epic: moderate (dust + some ingots)
- Epic→Mythic: expensive (dust + rare materials like netherite scrap/echo shards)

This cost curve encourages the player to settle for “good enough” gear until they're ready to invest in perfect rolls.

Interdependency notes for Wave 3

- **Spell Engine** is required by both Relics RPG and Skill Tree (RPG Series) — it was previously considered for removal but must be retained as a mandatory dependency of these core RPG mods.
- **Ranged Weapon API** is shared between Relics RPG and Skill Tree — install once, used by both.

Every weapon rolls with random affixes, every accessory provides a unique ability, and a skill tree gives every action meaning. This wave makes you **feel** your character grow alongside your tech.

Skill Tree (RPG Series) — Character Growth

Every action earns XP. Spend points on permanent stat bonuses.

Getting started:

1. Controls → Skill Tree → set a key (**P** recommended)
2. Press key → skill tree GUI opens
3. Spend first points on **Strength** (+melee damage, +mining speed) or **Vitality** (+max HP, +regen)

Attribute	Effect	Best for
Strength	+melee damage, +mining speed	Everyone
Vitality	+max HP, +natural regen	Everyone
Agility	+movement speed, +attack speed	Combat, kiting
Magic	+spell damage, +mana regen	If using spell mods
Defense	+armor toughness, +damage reduction	Tank builds

How XP works: Every ore mined, mob killed, item crafted, crop harvested generates XP. Scales with action complexity.

Pufferfish's Skills: The underlying framework powering Skill Tree. Handles XP tracking and attribute modification behind the scenes. Auto-installed dependency — no separate config needed.

Tip: Set the keybind and start earning XP from minute 1. Lowest-investment, highest-return mod in this wave.

Simply Swords — Unique Weapons

60+ unique weapons with passive and activated abilities. Drop from mobs (~2% chance), chests, and bosses.

Category	Examples	Playstyle
Heavy	Greathammer, Greatsword, Glaive	Slow, high damage, knockback. Two-handed
Medium	Longsword, Katana, Claymore	Balanced speed/damage. Versatile
Light	Rapier, Dagger, Sai, Cutlass	Fast, low damage per hit. Armor penetration on rapiers
Polearms	Spear, Halberd, Pike	Extended reach (2-3 blocks). Throwable spears
Special	Scythe (AoE), Soul Harvester (lifesteal)	Unique mechanics

Weapon abilities (hold **Ctrl** while hovering to see):

Ability	Effect	Found on
Whirlwind	Spin attack, hits all nearby	Greathammer, Greatsword
Impale	Charge + release for piercing thrust	Spear, Pike
Backstab	+300% damage from behind	Dagger, Sai
Lifesteal	Heal 10% of damage dealt	Soul Harvester, Scythe
Execute	+50% damage to targets below 30% HP	Axe variants

Finding weapons:

- Mob drops: ~2% chance. Zombies = common, skeletons = ranged-adjacent
- Chests: Dungeon/mineshaft/YUNG's structure chests have higher rates
- Trading: Some from weaponsmith villagers
- Bosses: Higher guaranteed drops from Twilight Forest and Cataclysm bosses

Too Many Bows — Unique Ranged Weapons

30+ unique bows with special abilities and custom attributes.

How they work: Each bow has a unique passive or activated ability. Check JEI for recipes and stats. Some shoot multiple arrows, some inflict status effects, some have special draw mechanics.

Finding them: Dungeon chests, mob drops, and boss loot. Combine with Apotheosis gem socketing for even more damage.

Integration: Simply Swords + Too Many Bows covers melee and ranged. Better Combat (Wave 5) provides appropriate animations for both.

Spartan Weaponry Unofficial — Polearms & Throwing

Spears, halberds, pikes, lances, throwing knives — fills the polearm gap that Simply Swords doesn't fully cover.

Key feature: Extended reach on all polearms (hit from 2-4 blocks away). Throwing weapons return after being thrown.

Finding them: Crafted at table (check JEI — wood/stone/iron/gold/diamond/netherite tiers). Lower tiers craftable early; higher tiers gate behind material progression.

Relics RPG — Legendary Accessories

Unique accessories for **Curios slots**. You have 6 slots: Ring (2), Amulet (1), Belt (1), Gloves (1), Back (1), Charm (1). Press Curios keybind (Y recommended) to open inventory.

Relic	Slot	Ability	Where to find
Monkey Talisman	Charm	Double jump	Vanilla dungeon chests
Holy Water	Charm	Right-click cleanses negative effects	Desert temples, churches
Lucky Coin	Charm	+luck for better loot	Mineshafts, outposts
Captain's Hook	Gloves	Pull entities toward you	Shipwrecks, ocean ruins
Sacred Wardstone	Amulet	Absorbs X damage then breaks	Stronghold chests
Magic Mirror	Back	Teleport to spawn/home	End cities, libraries
Heart of the Void	Ring	+max HP, slow regen	Nether fortresses, bastions
Blazing Feather	Back	+15% movement speed	Jungle temples, pagodas
Drowned Belt	Belt	Underwater speed boost	Ocean monuments, shipwrecks

Stacking: Two rings = two ring relics. Wear multiple different types.

Finding strategy: Use Explorer's Compass → find YUNG's structures → loot them. Relics have higher drop rates in YUNG's structures than vanilla ones.

Runes — Socketable Weapon Modifiers

Swapable modifications for weapons/tools. Change them between fights.

How to get:

- Loot chests in any structure
- Craft at **Rune Altar** (4 cobblestone 2×2 + lapis block on top, right-click with rune fragment)
- Low drop chance from hostile mobs

Rune	Effect	Best on
Fire	On-hit fire damage (3 sec)	Fast weapons (daggers, rapiers)
Ice	On-hit slow (2 sec)	Kiting weapons (spears, scythes)
Lightning	Chance to strike lightning	Heavy weapons (single big hits)
Poison	On-hit poison (5 sec)	DoT builds, boss kiting
Lifesteal	Heal 5% of damage dealt	Tank builds, slow weapons
Critical	+10% crit chance	Medium weapons

Using runes:

1. Craft **Rune Pouch** (leather + string + lapis) — stores all runes
2. Place weapon + rune at **Rune Altar** (4 stone + 1 lapis block)
3. Rune applied. To swap: place weapon back with different rune

Tip: Keep 2-3 loadouts. Fire for general combat, Lifesteal for bosses, Ice for kiting.

Apotheosis — Loot & Enchanting Overhaul

The single largest change to how gear feels.

Rarity System

Every item drop rolls with rarity:

Rarity	Affix slots	Color	Source
Common	0	White	Basic mobs, mining
Uncommon	1	Yellow	Hostile mobs, basic chests
Rare	2	Cyan	Dungeon chests, mini-bosses
Epic	3	Purple	Bosses, late-game chests
Mythic	4	Gold	Endgame bosses, Cataclysm

Higher rarities = more gem sockets + better affixes.

Affixes — Random Stats

Affix	Effect	Gear type
Ruthless	+crit damage, +crit chance	Melee weapons
Arcane	+spell damage, +mana	All weapons
Sturdy	+armor, +knockback resist	Armor
Nimble	+movement speed, +attack speed	Boots, leggings
Plentiful	+mining speed, +fortune	Tools
Titanic	+max HP, +natural regen	Chestplate

Tip: Check every drop. A Common helmet with a good affix can beat an Epic with useless stats. Salvage unwanted gear at a salvaging table for gem dust.

Gem System

Socketable permanent stat boosts:

Tier	How	Effect
Rough	Mob/chest drops	Basic stat boost
Cut	Rough + Gem Cutting Station + dust	+50% stat value
Flawless	2 Cut gems of same type + station	+100% stat value

Gem Cutting Station: Craft (check JEI), place rough gem + dust → cut gem. Dust from breaking unwanted cut gems.

Gem Socketing Table: Craft (check JEI), place item + gem in socket slot. Limited sockets per gear piece — expand with socketing tools.

Enchanting Overhaul

- **Higher level cap:** Sharpness VIII, Protection VIII, Fortune V, Looting V possible
- **Tiered bookshelves:** Hell, Seashelf, Endshelf enable higher enchantment levels
- **Library of Alexandria:** Stores all enchanted books in one block
- **30+ new enchantments:** Check JEI for full list

Boss Affixes

Boss mobs spawn with modifiers under their health bar:

Modifier	Effect	Strategy
Tough	+50% HP, damage reduction	High-damage weapons + Lifesteal
Blinding	Blinds on hit	Blindness-resistant gear or milk
Clumsy	Disarms on hit	Speed — don't get hit
Plagued	Poison/wither on hit	Antidotes, regeneration
Huge	2× size, more knockback	Stay mobile
Chaotic	Random effects per hit	Unpredictable — bring all potions

Tip: Boss affixes are the main source of Mythic gear. Hunt affixed bosses for endgame upgrades.

Spawner Module

Silk-touch a spawner with a pickaxe → modify it (entity type, spawn delay, player range, min/max entities). Right-click placed spawner with empty hand to fine-tune. Essential for XP farms and mob automation.

Deadly Module

Enabled. Adds telegraphed death-traps to the world (spikes, poison darts, instakill rooms, lava traps). Deadliest rooms hold the best loot. Traps are avoidable with caution. GraveStone Mod preserves your items if you trigger one.

Session Plan — Wave 3

Session	Focus	What to Do
Session 1 — Skill tree	Set keybind (P). Spend first points on Strength or Vitality	Permanent progression started
Session 2 — First affix loot	Check every gear drop. Look for Rare+ with useful affixes	Learn the loot system
Session 3 — Weapon hunt	Explorer's Compass → YUNG's dungeon → loot for Simply Swords weapon. Try Rapier (fast) or Greathammer (AoE)	First unique weapon
Session 4 — Gem cutting	Collect rough gems. Build Gem Cutting Station. Cut Rough → Cut. Socket into gear	Permanent gear upgrade
Session 5 — Relic hunting	Monkey Talisman (dungeons) + Lucky Coin (mineshafts)	Accessory slots filled
Session 6 — Boss affixes	Search for affixed bosses (colored name above health bar). Best source of Mythic gear	Endgame gear progression
Session 7 — Rune loadout	Craft Rune Pouch + Rune Altar. Apply Fire rune to fast weapon, Lifesteal to slow weapon	Swappable combat modifiers

Tip: Wave 3 gear changes how every other wave feels. A Rare+ weapon with the right affix makes Twilight Forest and Cataclysm bosses dramatically easier. Invest in gear before tackling tough content.

Wave 4 — Food, Farming & Comfort

Mod	Role
Farmer's Delight 1.3.2	Cooking hub — cutting board, cooking pot, stove, new crops
Brewin' And Chewin' 4.5.0	Drinks and fermentation — beer, wine, mead, whiskey
Storage Delight 1.3.0	Kitchen furniture — drawers, cabinets, shelves for ingredient storage
More Delight 26.04.18a+1.21-neoforge	Extra meals, sandwiches, toasts, and knife variants
My Nether's Delight 1.10.2	Nether-themed food ingredients and recipes
Ocean's Delight 1.0.4	Ocean-themed seafood dishes
Ender's Delight v1.3.0	End-themed food for Farmer's Delight (moved from Wave 2)
Neo Bee Fix	Fixes bee AI bugs — bees leave hives, don't fall from sky, no NW wander bias
Comfortable Campfires	Campfires grant configurable status effects — sit by the fire, get a buff

Farmer's Delight brings a full cooking ecosystem. Optional — you can survive on baked potatoes — but cooking gives powerful status effects, better healing, and a cozy kitchen to build around.

Farmer's Delight — Cooking Hub

New Crops

Crop	Grows on	Use
Tomato	Farmland. Bush grows, pick when red	Soups, sandwiches, salads
Cabbage	Farmland. Full head when ready	Coleslaw, cabbage wraps
Onion	Farmland. Pick when shoots show	Savory dishes, soups
Rice	Shallow water (1 block deep on dirt/mud)	Hearty meals, sushi

Wild crops: Break wild variants in the world for seeds:

- Wild tomatoes — plains/forests
- Wild cabbage — beaches
- Wild onions — taiga
- Wild rice — swamps

Cutting Board

Craft: 1 knife + 2 planks (horizontal row). Place on ground, right-click to open GUI.

Place ingredient in left slot → processed output in right slot.

Input	Output	Use
Cooked meat	Meat strips	Sandwiches, salads
Cabbage	Cabbage leaves	Wraps, coleslaw
Bread	Bread slices	Toast, sandwiches

Input		Output		Use	
Carrot		Carrot slices		Salads, soups	
Apple		Apple slices		Desserts, snacks	

Tip: Knife in off-hand while chopping lasts longer (uses knife durability). Iron knife = dozens of chops.

Cooking Pot

Craft: 2 iron + 2 planks + 1 bowl. Place over a heat source:

- **Stove** (fastest) — 4 bricks + 4 iron nuggets
- **Campfire** (medium) — 3 logs + 3 sticks + 1 coal
- **Lava** (slowest) — pot must sit directly above lava source

Cooking process:

1. Right-click pot → open GUI
2. Fill water slot (bucket or bottle)
3. Add up to 6 ingredients in any order
4. Wait for progress bar
5. Collect meal — most give **6 servings** with bowl stacking

Meal categories:

- **Soups** — Tomato soup, beef stew. Good saturation, easy ingredients
- **Sandwiches** — Chicken sandwich, BLT. On-the-go eating
- **Salads** — Mixed salad, fruit salad. Good for hot biomes
- **Pasta** — Pasta with meat sauce. Filling, complex ingredients
- **Rice meals** — Fried rice. Requires rice farming

Kitchen Storage

Craft **oak/pantry cabinets** (check JEI — varies by wood). Function as chests with kitchen aesthetics.

Tip: Set up kitchen near your AE2 network. Cook manually or stock ingredients in ME drives for autocrafting.

Brewin' And Chewin' — Drinks & Fermentation

Alcohol brewing with aging mechanics. Grow hops, barley, grapes on farmland.

Brewing Station

Craft (check JEI — barrel-like recipe). Right-click to open GUI.

1. Add **water** + main ingredient:
 - Barley → beer base (wort)
 - Grapes → wine base (must)
 - Honey → mead base
 - Potatoes → spirits base (mash)
2. Wait for fermentation
3. Collect **fermented mixture**

Aging Barrel

Craft (check JEI — wooden barrel). Place fermented mixture inside, right-click to seal.

Ages one stage per in-game day: Young → Matured → Aged → Vintage (4 stages). Each stage increases status effect potency and duration.

Optimal: Let everything reach **Aged** (stage 3). Push to Vintage (stage 4) only for special occasions.

Drink	Matured effect	Vintage effect
Beer	Resistance I (2 min)	Resistance II (5 min)
Wine	Haste I (2 min)	Haste II (4 min)
Mead	Speed I (2 min)	Speed II (4 min)
Whiskey	Strength I (30 sec)	Strength II (1 min)
Vodka	Night Vision (3 min)	Night Vision + Fire Resist (5 min)

Practical use: Brew a batch of each, age to Matured/Aged, store in kitchen. Pop a drink before boss fights for free buffs. Mead stacks with Swiftess potions.

Bottle return: Empty bottles are returned after drinking. Reuse at brewing station.

Storage Delight — Kitchen Furniture

Decorative kitchen storage that works as functional inventory:

Block	Slots	Use
Cabinet	27	Counter-height storage
Drawer	18	Fits under counters
Shelf	9	Wall-mounted spice display

Tip: Use for early-game ingredient storage before AE2. Later, place an ME Storage Bus on a cabinet to bridge into your digital network.

More Delight — Extra Meals & Knives

- **Extra meals:** Sandwiches, toasts, stuffed mushrooms
- **Knife variants:** Diamond and netherite knives (much longer lasting than iron)
- **New cooking pot recipes:** Stews, soups, pasta dishes

All recipes visible in JEI. Uses existing cutting board and cooking pot.

My Nether's Delight — Nether Food

Nether-themed ingredients and recipes:

- **Nether crops:** Grow in soul sand or warped/crimson nylium
- **Ingredients:** Glowstone berries, soul berries, nether-root vegetables
- **Recipes:** Spicy stews (Fire Resistance), soul-force snacks (Night Vision)

Tip: Set up a Nether farm for combat-relevant status effects. Glow berry pie gives Haste II — useful before mining expeditions.

Ocean's Delight — Seafood

Seafood dishes using ocean mob drops:

- **Ingredients:** Squid tentacles, fish fillets, seaweed, turtle eggs
- **Recipes:** Seafood stew, grilled fish, kelp salads, sushi rolls
- **Effects:** Water Breathing + Dolphin's Grace — perfect for ocean monument raids

Tip: Build a fish farm or farm kelp for steady supply. Combined with My Nether's Delight, prepare meals for any environment.

Ender's Delight — End Food

End-themed food for post-Dragon exploration:

- **Ingredients:** Chorus fruit variants, end-themed crops from BetterEnd
- **Recipes:** End stews and snacks
- **Effects:** Status boosts useful for End dimension exploration

Integration: End ingredients from BetterEnd biomes (Wave 2) cook into Ender's Delight meals. Plan an End farming outpost with Farmer's Delight crops.

Neo Bee Fix

Fixes bee AI — bees properly leave hives, don't fall from sky, no northwest wander bias. No config. Install and forget.

Integration: Bees produce honey for Brewin' And Chewin' mead. Farm bees near your kitchen for renewable brewing ingredients.

Wave 4.5 — Quests

Procedural bounty system that gives purpose to exploration and material collection without manual quest authoring.

Mod	Role
Bountiful 8.0.0-beta.2	Bounty boards in villages — procedurally generated fetch/kill/exploration quests with rewards

Bountiful generates quests automatically — find a bounty board in any village, accept tasks, earn rewards. Fully optional. Items from installed mods automatically appear in its reward pools.

Procedural bounty boards give you optional goals as you explore. No quest book, no authoring, no pressure.

Bountiful — Bounty Boards

Find **bounty boards** in every village (lectern-like block near center). Right-click → quest list.

Quest Type	Example	Reward
Gathering	Deliver 16 iron ingots	Emeralds, XP, rare items
Hunting	Kill 8 creepers	Combat loot, XP
Exploration	Find a specific biome	Map items, navigation gear
Crafting	Deliver crafted tools/armor	Ingots, gems

How:

1. Find any village
2. Locate bounty board
3. Right-click → select a quest
4. Required items appear in board's slots — drop them in
5. Collect reward
6. New quests regenerate periodically

Decree system: Blacklist quest types you don't enjoy.

Tip: Take gathering quests for materials you already have in your AE2 system — free rewards for zero extra work.

Session Plan — Wave 4 & 4.5

Session	Focus	What to Build
Session 1 — Kitchen tools	Cutting board + cooking pot + stove. Find wild crop seeds	Functional kitchen
Session 2 — Farm	Plant tomato, cabbage, onion, rice. Cook first meal (tomato soup is easiest)	Sustainable ingredients
Session 3 — Brewing	Find barley/grapes/honey. Build brewing station. Start first batch. Place aging barrel	First drink fermenting

Session		Focus		What to Build	
Session 4 — Kitchen expansion		Storage Delight cabinets/ drawers. Try More Delight recipes. Collect Nether ingredients		Full kitchen	
Session 5 — Bounties		Visit village, find bounty board, complete a gathering quest for free rewards		Passive XP and loot	
Ongoing		Age drinks for boss buffs. Explore oceans for seafood. Stock AE2 with ingredients for autocrafting		Buff food always ready	

Wave 5 — Combat & Mobs

Mod	Role
Better Combat	Melee combat animations — weapon reach, sweeping, attack speed
Mutant Monsters	Mutant zombie, skeleton, creeper, enderman — unique drops
L_Ender's Cataclysm	Epic boss fights — Leviathan, Netherite Monstrosity, Ender Guardian (CurseForge, NeoForge 1.21.1)
When Dungeons Arise	Roguelike combat structures with loot
Enchantment Descriptions	Shows enchantment effects in tooltips
Create Big Cannons	Artillery engineering — siege cannons for colony defense and spectacle
Cut Through	Combat QoL — swing through transparent blocks (tall grass, cobwebs) without breaking them

Apotheosis also adds boss modifiers and elite mob affixes — partially covers what Champions would have done.

Your powerful gear finally has worthy opponents. Better combat mechanics, tougher mobs, epic boss fights, roguelike dungeons.

Better Combat — Weapon Feel

Replaces Minecraft's single swing with weapon-specific movesets:

Weapon	Speed	Reach	Animation	Best for
Daggers	Very fast	2 blocks	Quick stab	Debuff stacking, fast crit
Swords	Fast	3 blocks	Horizontal arc	General combat
Rapiers	Fast	3.5 blocks	Thrust (ignore shields)	Shield-breaking
Greatswords	Slow	4 blocks	Wide sweep	Group clearing
Axes	Medium	3 blocks	Overhead chop	Armor penetration
Spears	Medium	4.5 blocks	Forward thrust	Kiting
Hammers	Slow	3.5 blocks	Overhead slam	Knockback, stun
Greathammers	Very slow	4 blocks	Ground pound	AoE knockback

Config: Mod Menu → Better Combat. Tweak reach, sweep radius, attack speed, visual effects per weapon type.

Integration: Simply Swords weapons have built-in Better Combat data files — each category uses correct animation, reach, and speed automatically. Too Many Bows and Spartan Weaponry also integrate.

Enchantment Descriptions

Hover over any enchantment in tooltip → detailed description of what it does. Works in enchanting table GUI, on items, and in books. No config.

Cut Through — Clean Combat

Swing through transparent blocks (tall grass, vines, cobwebs) without breaking them. Hit the mob behind. No more breaking your grass farm or getting stuck on cobwebs while skeletons shoot you.

No config. Client-side only. Works with Better Combat and all other combat mods.

Mutant Monsters — Boss Variants

Tougher versions of vanilla mobs. Drop unique gear.

Mutant	Spawns	What to expect	Drops
Mutant Zombie	Overworld, night	Tankier, ground-pound AoE, summons zombies	Hulk Hammer (AoE knockback)
Mutant Skeleton	Overworld, night	Rapid-fire weakness arrows, teleports	Mutant Skeleton Armor (projectile resist)
Mutant Creeper	Overworld, night	2× explosion radius, drops mini-creepers on death	Creeper Minion (follows, explodes on command)
Mutant Enderman	The End	Aggressive teleporting, throws blocks, projectile resist	Enderman gear (teleportation items)

Spawn rate: Adjust in Mod Menu → Mutant Monsters → spawn weight if too frequent/rare.

Strategy: Same as vanilla but more extreme. Zombie → kite. Skeleton → close fast. Creeper → ranged. Enderman → water + fire resistance.

L_Ender's Cataclysm — Epic Boss Fights

Hand-animated bosses with unique mechanics. **Opt-in** — you must build a summoning structure or find a specific location.

Boss	Location	How to Summon	HP	Recommended Gear
Leviathan	Deep ocean (Y<40)	Prismarine + sea lantern altar + Prismarine Shard	~200	Diamond, Depth Strider, door trick
Netherite Monstrosity	Nether wastes	Enter central arena in nether brick lair	~300	Full diamond/netherite, fire resistance
Ender Guardian	End (outer islands)	Eye of Ender on pedestal after Dragon	~400	Endgame gear, Elytra + Slow Falling
Ancient Remnant	Desert	Enter summoning room in large pyramid, insert Nether Star	~350	Blast protection, blocks to pillar
Ignite	Nether (basalt deltas)	5×5 hollow obsidian sphere +	~250	Fire resistance, water buckets

Boss	Location	How to Summon	HP	Recommended Gear
		Blaze Rod center, wait 10 sec		

General strategy:

1. Scout the arena first
2. Bring food buffs (Farmer's Delight + Brewin' And Chewin' stack)
3. Apply runes (Lifesteal or Ice)
4. Set a Waystone nearby
5. AE2 wireless terminal for mid-fight resupply

Leviathan: Phase 1 — dodge charges, attack body, jump shockwaves. Phase 2 (50%) — avoid whirlpools, attack head between spawns. Drops Leviathan Blade (trident-like water weapon).

Netherite Monstrosity: Phase 1 — attack legs to slow. Phase 2 (60%) — enrages, faster attacks + fire breath. Sprint behind pillars. Drops Monstrosity tools (efficient netherite-level mining).

Ender Guardian: Phase 1 — dodge tracking energy bolts, hit wall crystals to stun. Phase 2 (50%) — teleports, summons endermite, break all crystals to lower shield. Phase 3 (20%) — energy nova, pillar up to avoid ground AOE's. Drops high-damage weapons + teleportation gear.

Ancient Remnant: Phase 1 — sandstorm blinds, attack during cooldown. Phase 2 (40%) — sinks into sand, pops up elsewhere, shoot exposed weak point. Drops tools with unique enchantability.

Ignite: Phase 1 — fireball barrage, block with shield or dodge behind pillars. Phase 2 (40%) — absorbs lava from arena, stay on high ground, attack when emerging. Drops fire weapons (permanent fire aspect).

When Dungeons Arise — Roguelike Structures

Large multi-floor combat structures across the Overworld.

Structure	Floors	Difficulty	Biome	Loot
Abandoned Temple	3	Easy	Plains, forests	Iron gear, early rare items
Small Castle	4	Medium	Plains, hills	Rare weapons, gold
Large Castle	6	Hard	Mountains	Epic gear, diamonds
Barracks	2	Easy	Any	Armor, shields
Scorched Fort	4	Medium	Desert, badlands	Fire-related items
Warped Fortress	5	Hard	Any (rare)	Endgame gear, unique loot
Foundry	3	Medium	Underground	Ores, mining gear
Bandit Camp	1	Easy	Forest, taiga	Early weapons, food

Finding: Explorer's Compass → "When Dungeons Arise" filter → select structure.

Strategy: Clear floors systematically. Loot scales with depth. Bring blocks to pillar, torches to light hallways, decent gear for deep floors.

Note: Structures are finite — once looted, they don't respawn unless chunks reset. Mark looted ones on JourneyMap.

Create Big Cannons — Artillery Engineering

Functional cannons built from Create materials, using rotational power for assembly and gunpowder for ammo.

Getting started:

1. Craft cannon components: barrel (bronze/steel), mount (Create materials), charger
2. Build: place mount, attach barrel, connect charger
3. Arm: insert powder charges + cannonballs into GUI
4. Fire: right-click or redstone signal

Ammo	Effect	Use
Solid cannonball	High damage, block-breaking explosive	Siege, defense
Canister shot	Spread of small projectiles	Anti-personnel
Grape shot	Medium spread	Crowd control
Incendiary	Fire on impact + lingering fire	Area denial
Smoke shell	Smoke screen	Concealment

Integration: Gated behind bronze/steel (Create progression) + gunpowder. Mount on colony walls for defense. Redstone-controlled pulse extender for automatic salvo fire.

Dependency: Ritchie's Projectile Library (auto-installed) — provides custom trajectories and hit detection. No separate config.

Session Plan — Wave 5

Session	Focus	What to Do
Session 1 — Combat feel	Better Combat works automatically. Craft/find a Simply Swords weapon. Test daggers (fast stab) vs greatswords (wide sweep). Equip a rune	Learn your weapon
Session 2 — First mutant	Find Mutant Zombie at night (listen for roar). Kite it, learn patterns. Collect unique drop	First mini-boss
Session 3 — First dungeon	Explorer's Compass → When Dungeons Arise small castle/abandoned temple. Clear floors	Roguelike dungeon complete
Session 4 — First Cataclysm boss	Easiest = Leviathan (deep ocean) or Netherite Monstrosity (Nether). Prepare: good weapons, fire resistance, healing food	Epic boss beaten
Session 5 — Cannons	Craft bronze (zinc + copper in heated mixer). Build Small	Artillery online

Session		Focus		What to Do	
		Cannon. Test fire. Mount near base entrance			
Session 6+ — Boss gauntlet		Remaining Cataclysm bosses. Each drops unique endgame materials. Revisit Twilight Forest bosses with Wave 5 gear		Full endgame gear	

Tip: Better Combat changes melee fundamentally. Don't skip Session 1 even as a ranged player.

Wave 6 — Building & Decoration

Core Decoration

Mod	Role
Rechiseled	600+ decorative block variants with connected textures
Supplementaries	Vanilla+ decoration — jars, signposts, faucets, weather vanes
Macaw's Bridges	Bridges in all wood types
Macaw's Doors	Vanilla-style doors for every wood
Macaw's Roofs	Roofing blocks
Macaw's Windows	Window variants
Building Wands	Faster large-scale construction
[Reforged] Accurate Block Placement	Precision block placement — place on the exact face you're looking at
Handcrafted	Furniture — chairs, tables, shelves, curtains
The Magic Mirror	Reflective mirror block that shows the nearest player's reflection

Rechiseled Addons

Mod	Role
Rechiseled: Create	100+ decorative variants of Create blocks + mechanical chisel
Rechiseled: Applied Energistics 2	AE2 autocrafting for Rechiseled blocks via Chiseling Pattern Encoder

Tools and blocks to build the base you deserve.

Rechiseled — Decorative Block Variants

600+ decorative block variants with connected textures.

How:

1. Craft **chisel** (iron + 2 sticks, check JEI)
2. Right-click with chisel → chiseling screen
3. Place base block in input slot → browse all variants
4. Click variant to preview (cycles through single block, row, 3×3 grid)
5. Take result from output slot
6. **Left-click** placed blocks with chisel to cycle through random variants in-world

Key features:

- **Connected textures** — toggle with bottom-right button
- **Stairs & slabs** — every variant has stair/slab forms
- **No durability** — chisel never breaks

Block types (30+ variants each):

- Stone (bricks, tiles, carved, pillars, panels, runed, mosaic)

- Glass (all colors + connected clear, borders, panels)
- Wood (grain variants, carved panels, ornate borders)
- Wool & Carpet (woven, knitted, cross-stitched)
- Concrete (smooth, paneled, tiles, bordered)
- Nether materials (glowstone, nether bricks, blackstone)

Addons:

- **Rechiseled: Create** — 100+ decorative Create block variants + mechanical chisel
- **Rechiseled: AE2** — autocraft Rechiseled blocks via AE2's Chiseling Pattern Encoder

Supplementaries — Vanilla+ Functional Decoration

Decoration blocks that serve a purpose:

Block	Function	How
Jar	Stores 1 item visually	Right-click with item to put in, empty hand to take out
Signpost	Directional sign pointing to named location	Right-click with tag item to set destination name. Arm points toward Waystone
Faucet	Fluid flow decoration	Place above fluid tank. Right-click to toggle flow
Weather Vane	Shows wind direction/speed	Place outside. Rotates in real-time
Item Shelf	Wall-mounted display (3 items)	Right-click with item to place, sneak-right-click to remove
Rope	Climbable, place in mid-air	Right-click to place knot, add more rope below. Breaks when top knot broken
Flute	Craftable instrument	Right-click to play note. Villagers react

Practical builds:

- Kitchen: Item shelves for ingredients, jars for flour/sugar, faucet over sink
- Entrance: Signposts to "Mine", "Nether Portal", "Village"
- Garden: Weather vane, rope swing from tree
- Study: Jars with glowing items as lamps

Macaw's — Complete Building Set

Five modules covering structural elements in every wood type.

Bridges: 6 slabs + 1 stick in bridge pattern. Right-click to extend, shift-right-click to shorten. Straight or rope variants. All 10+ wood types.

Doors: Special doors per wood — glass, slatted, stable (top half opens independently), cottage, shop, paper. Craft with matching wood + relevant material.

Roofs: Roofing blocks — slopes, ridges, gutters. 3 slabs in triangle pattern for slope piece. Auto-connects corners.

Windows: Diamond pane, framed, barred, tropical, stained glass. Craft with glass + wood.

Fences: Picket, stockade, horse, wired, highley gate. Each distinct look.

Building tip: Macaw's roofs + Rechiseled stone + Macaw's windows = cohesive aesthetic.

Building Wands — Large-Scale Construction

Place many blocks at once. Closest thing to WorldEdit without a server plugin.

Craft: Stick + 2 blocks of same type in + pattern. Block type determines tier (stone = basic, iron = better, diamond = best).

Modes (right-click wand to open GUI):

Mode	Does	When to use
Build Wall	Wall between 2 clicked points	Enclosing rooms, perimeters
Build Line	1-block-wide line	Railings, borders, paths
Fill Area	3D cuboid between 2 corners	Foundations, clearing rooms
Build Circle	Ring of blocks	Towers, wells
Build Cylinder	Filled cylinder	Silos, water tanks
Build Sphere	Hollow or filled sphere	Domes, decorative globes

How: Left-click to set first corner (green sparkle). Right-click to set second corner (structure builds). Uses blocks from your inventory.

Tips:

- Shift+Right-click to undo last placement
- Use **Undo wand** (stick + slimeball) to reverse recent actions
- Combine with AE2 wireless terminal for inventory access while building

Handcrafted — Furniture

Furniture	Function	Craft
Chairs	Right-click to sit	Slabs + planks
Tables	Right-click to place items on top	Varies by type
Shelves	Wall-mounted item display	Wood + relevant material
Curtains	Right-click to open/close	Wool + stick
Desks	Table + drawer storage	Slabs + planks
Cupboards	Wall-mounted aesthetic storage	Wood + glass
Sofa	Multi-block seating	Wool + wood
Dresser	Tall drawer storage	Wood + relevant material

Tip: Place in your most-used rooms — kitchen, dining, bedroom. Use shelves to display favorite loot (Twilight Forest trophies, Apotheosis mythic gear, rare Simply Swords weapons).

The Magic Mirror

Reflective mirror block showing the nearest player's reflection.

How: Craft (check JEI — glass + frame materials). Place on wall. Shows your character model in real-time, including armor and held items.

Use: Decorative vanity mirrors in bedrooms and dressing rooms. Purely aesthetic — no functional gameplay effect.

Session Plan — Wave 6

Session		Focus		What to Build	
Session 1 — Rechiseled basics		Craft chisel. Convert stone, wood, glass into decorative variants. Try connected textures		Decorative block library	
Session 2 — Structural elements		Craft Macaw's roofs, windows, doors in your base wood type. Build a cohesive room		Cohesive room aesthetic	
Session 3 — Furniture		Handcrafted chairs, tables, shelves in kitchen and bedroom. Item shelves for loot display		Functional interior	
Session 4 — Large-scale		Building Wands — wall mode for perimeter, fill mode for foundation. Build a tower with circle mode		Rapid construction	
Session 5 — Details		Supplementaries jars, signposts, weather vanes. The Magic Mirror in bedroom. Rechiseled: Create for factory aesthetics		Decorative details	

Wave 7 — Space Exploration

Core Space Mod

Mod	Role
Stellaris 1.4.23	Space exploration — rockets, Moon/Mars/Venus/Mercury/Glacio, oxygen system, machines, energy

Stellaris (by ExodusTeam) is the successor to Beyond Earth and the premier space mod for NeoForge 1.21.1. It features 4 rocket tiers, a rover, oxygen management, custom gravity per planet, oil/water/fuel processing, radioactives, and alien mobs. Machines include the Oxygen Distributor, Water Separator, Solar Panel, Fuel Refinery, and Pump Jack. 250+ building blocks across planet-specific sets.

Integration with Tech Mods

Layer	Stellaris ↔ Tech Mod Interaction
Power	Mekanism cables can power Stellaris machines. Create rotational → Mekanism → Stellaris
Energy	Mekanism universal cable connects to Stellaris machines (energy API compat)
Storage	AE2 ME drives store Stellaris materials; autocrafting builds rocket components
Transport	Create trains move space resources from landing pad to base. Sophisticated Backpacks carry oxygen tanks
Ore Processing	Mekanism 5x ore multiplication works on Stellaris ores (desh, ostranium, etc.)
Food	Farmer's Delight meals eatable in space when oxygen is present (v1.2.1+)
Gear	Apotheosis affixes may apply to Stellaris space suit (if item tags align)

Progression Placement

Phase 2 — Industrial Ascent (hours 40–200). Rockets require advanced materials that gate them behind Mekanism mid-game processing and AE2 infrastructure. Oxygen systems and fuel refining add a parallel supply-chain puzzle that rewards industrial automation.

Mod Count Summary

Wave	Mods	Deps	Total	Notes
Wave 0 — Foundation	14	10	24	Dependencies (12), Performance & Rendering (10), Shaderpack (1 mod: Euphoria Patches), Infrastructure (3) — all the nuts & bolts that make the game run (+1 shaderpack, +10 resource packs)
Wave 0.5 — Player Experience	45	—	45	Visual & Client QoL (27), Inventory & UI (14), Storage & Travel (6), Utility & QoL (1: Easy

Wave		Mods	Deps	Total	Notes
					Anvils), Village & Trading (2) — everything that makes the game feel good
Wave 1 — Tech		18	4	22	Create + 12 addons (incl. Power Grid, Gunsmithing, Ornithopter Glider, Train Utilities, Railways Navigator, Interiors, Train Parts, Threaded Trains), Mekanism + Generators, AE2, Advanced Finders, Advanced Chimneys + NTGL, GeckoLib, Framework, ForgeEndertech deps
Wave 1.5 — Colony		5	—	5	MineColonies (4 deps counted as mods — they're library mods; CurseForge-only)
Wave 2 — Exploration		24	6	30	YUNG's (9), Terralith/Tectonic, Serene Seasons, Darker Depths, Upgrade Aquatic, dimensions (3), End overhaul (2 + Nullscape dp + 6 deps), navigation, aircraft, ships
Wave 3 — Equipment Magic		9	10	19	Skill Tree, Simply Swords, Too Many Bows, Spartan Weaponry, Relics, Runes, Curios, Apotheosis + 10 deps (Placebo, Apothic modules, Patchouli, Simply Tooltips, Fzzy Config, Ranged Weapon API, Spell Engine, Bundle API)
Wave 4 — Food & Farming		9	—	9	Farmer's Delight + 6 addon mods + Neo Bee Fix + Comfortable Campfires
Wave 4.5 — Quests		1	—	1	Bountiful
Wave 5 — Combat		7	6	13	Better Combat, Mutant Monsters, Cataclysm, Dungeons Arise, Big Cannons, Cut Through, Enchantment Descriptions + RPL lib, playerAnimator, Puzzles Lib, Citadel, Lionfish-API, Bookshelf
Wave 6 — Building		11	3	14	Rechiseled, Supplementaries, Macaw's (4), Building Wands, Reforged: Accurate Block Placement, Handcrafted + Rechiseled: Create, Rechiseled: AE2 + Moonlight Lib, Resourceful Lib, Fusion
Wave 7 — Space Exploration		1	1	2	Stellaris, Potentials API

Wave		Mods	Deps	Total	Notes	
Total		144	40	184	All confirmed NeoForge 1.21.1	

Space travel becomes real. Build rockets, manage oxygen, explore alien worlds, bring back exotic resources.

Stellaris — Interplanetary Travel

Complete space exploration mod. Build rockets piece by piece, manage oxygen and fuel, travel to 5 planets.

Video tutorial: [Stellaris Getting Started Guide](#)

Core concepts:

Concept	What it means
Rocket Components	Build engines, fuel tanks, capsules, fins separately → assemble at Rocket Station
Oxygen	Most planets have no atmosphere. Space suit + oxygen required
Fuel	Rockets need fuel (oil → refinery). Higher-tier rockets need more fuel
Planet Tiers	Basic rocket = Moon only. Upgraded rockets reach Mercury, Venus, Mars, Glacio
Gravity	Each planet has different gravity. Low = higher jumps. High = weighted movement

Phase 1: Getting to Space

Step 1 — Gather resources:

- Find **oil** in Overworld (pools in deserts/plains)
- Mine **steel** (from Mechanism Metallurgic Infuser or TFMG Blast Furnace)
- Collect iron, copper, redstone, glass

Step 2 — Build basic machines:

Machine	Function	Priority
Rocket Station	Assembly block — build components here	First
Fuel Refinery	Oil → rocket fuel	Second
Oxygen Distributor	Fills space suits with oxygen	Third
Water Separator	Water → oxygen (for Distributor)	Fourth
Solar Panel	Power generation on planets	Fifth

Step 3 — Build first rocket:

1. Open Rocket Station GUI
2. Craft **Tier 1 Engine, Tier 1 Fuel Tank, Capsule, Fins**
3. Rocket Station assembles them into **Tier 1 Rocket**
4. Right-click assembled rocket to create launch pad

Step 4 — Prepare for launch:

1. Fill fuel tank from Refinery (fluid pipes or buckets)
2. Equip **Space Suit** (check JEI — steel + glass + rubber/leather)
3. Fill suit with oxygen at Oxygen Distributor

4. Stock supplies: food (Farmer's Delight meals work with oxygen), blocks, torches, weapons

Step 5 — Launch:

1. Enter rocket (right-click door)
2. Press launch button in GUI
3. Watch ascent animation

The Moon — First Destination

Safest first destination. Low gravity (~17% Overworld), no atmosphere, basic resources.

What to do:

- Mine **moon stone** and **cheese ore** (smelts into cheese ingots)
- Collect **desh** — Moon's primary rare resource. Used for Tier 2 rocket components
- Explore structures — crashed spaceships, moon bases, landing pads
- Place a **Waystone** immediately
- Set up base: solar panel + oxygen distributor + chest
- Look for rich ore veins — higher concentrations of certain Overworld ores

Returning: Enter rocket on Moon → launch → returns to original launch pad.

Other Planets

Planet	Tier	Gravity	Hazards		Key Resource	Notes	
Moon	1	Low	None (oxygen required)		Desh	First destination. Safe practice	
Mercury	2	Low	Extreme heat, fire damage		Ostranium	Bring fire resistance	
Venus	2	High	Acid rain		Tharsite	Dense, heavy movement	
Mars	3	Medium	Cold, dust storms		Desh (rich)	Best for permanent off-world base	
Glacio	4	Low	Extreme cold		Glacio crystals	Ice planet. Endgame destination	

Off-World Machines

Machine		Function		Why you need it	
Solar Panel		Power from sunlight		Most planets have more sun than Earth	
Oxygen Distributor		Breachable air		Required on every planet	
Water Separator		Water → oxygen		Feed oxygen into Distributor	
Fuel Refinery		Oil → rocket fuel		Refine on destination planet if oil found	
Pump Jack		Extracts oil		Find oil pools on other planets	

Integration With Tech Base

System		How it connects	
Power		Mekanism generators at Moon base. Universal cables power Stellaris machines	

System	How it connects
Ore processing	Bring planet ores (desh, ostranium, tharsite) back to Mekanism line. 5x works on Stellaris ores
Storage	AE2 ME drives store Stellaris materials. Autocraft rocket components. Quantum link or P2P tunnel to off-world base
Transport	Create trains move rocket materials from launch pad to base. Conveyors feed Rocket Station
Fuel	TFMG diesel/gasoline for industrial use. Stellaris uses its own fuel crafting (oil → refinery)
Backpacks	Netherite-upgraded backpacks carry oxygen tanks and supplies
Food	Farmer's Delight meals eatable in space with oxygen present
Gear	Apotheosis affixes may apply to space suit (if item tags align)

Progression Placement

Space belongs in **Phase 2** (hours 40–200). Prerequisites:

- Mekanism basic processing (Enrichment Chamber)
- Small AE2 network
- Create power infrastructure
- Good weapons/armor (Wave 3/5 gear helps with planet mobs)

Start with Moon missions, establish Tier 1 base, then Mercury/Venus. Mars and Glacio are late Phase 2.

Session Plan — Wave 7

Session	Focus	What to Build
Session 1 — Ground prep	Find oil. Build Fuel Refinery + Rocket Station. Craft Tier 1 components. Assemble rocket	Launch infrastructure
Session 2 — Moon landing	Craft Space Suit. Fill with oxygen. Stock supplies. Launch	First off-world base, desh
Session 3 — Moon base	Solar panel + oxygen distributor. Waystone. Mine desh for Tier 2. Chunk Loader	Sustainable Moon operation
Session 4 — Next planets	Tier 2 rocket → Mercury or Venus. Unique hazards (heat/acid). Temporary base	New resources
Session 5+ — Deep space	Tier 3 → Mars (permanent base). Tier 4 → Glacio (endgame crystals). Digital Miner for auto-extraction	Interplanetary logistics

Tip: Space is the most resource-intensive activity. Don't rush it. Each planet should feel like a meaningful expansion. MekaSuit trivializes spacewalks — jetpack flight + environmental protection = no oxygen worries.

Keybindings Reference

Keybind		Suggested		Mod		Wave
View Recipes		R		JEI		0
View Uses		U		JEI		0
Open Fullscreen Map		J		JourneyMap		0
Toggle Minimap		Check default		JourneyMap		0
Open Waystone Menu		Ctrl+Shift+W		Waystones		0
Open Backpack		B		Sophisticated Backpacks		0
Zoom		C		Sodium		0
Toggle Dynamic Lights		Ctrl+L		LambDynamicLights		0
Open Curios Inventory		Y		Curios API		0
Open Skill Tree		P		Skill Tree		3
View Weapon Ability		Ctrl (hold)		Simply Swords		3
Open Rune Pouch		Set manually		Runes		3
Open Create Ponder		W (hold over block)		Create		1
Open Town Hall GUI		Right-click		MineColonies		1.5

Pro tip: Open **Options** → **Controls** → **Key Binds** and browse categories at the bottom. Every mod's binds are grouped under its name. Set the ones above first, then explore others as you install new waves.

Progression Through All Waves

Phase	What you'll be doing
Phase 1 (hours 0–40)	Wave 0 setup. Early Create (water wheel, millstone). Find certus quartz for AE2. First Mekanism machines (2x ore). Explore Terralith biomes. Skill Tree earning XP from first action. First Simply Swords weapon from dungeon. Cook first Farmer's Delight meal. Start MineColonies (Town Hall, Builder, Mine). Grab bounties from village boards
Phase 2 (hours 40–200)	Full Create factory (conveyors, deployers, sequenced crafters, trains). Mekanism 3x–5x ore processing. AE2 ME network with autocrafting. Explore Twilight Forest bosses. Apotheosis Rare/Epic gear. Mutant Monsters. First Cataclysm boss. TFMG oil refining and diesel engines. Build with Rechiseled, Macaw's, Handcrafted. Small Ships for ocean travel. Immersive Aircraft for aerial exploration. Complete bounties. Stellaris Moon and early planets
Phase 3 (hours 200+)	Mekanism fusion reactor + MekaSuit. AE2 full autocrafting CPU chains. Endgame Apotheosis Mythic gear with perfect gem sockets. Twilight Forest final bosses. Cataclysm endgame bosses. BetterEnd

Phase	What you'll be doing
	dimension exploration. Mars and Glacio space bases. Monumental building projects. Create mega-factories. Perpetual optimization

General Tips

Finding Help

- **JEI (R/U)** — always your first stop. Hover any item, press R for recipe, U for uses
- **Ponder (W over Create blocks)** — interactive animated tutorials. Better than any wiki
- **Mod Menu / Configured** — every installed mod's config screen
- Most mods link to their wiki or Discord from Mod Menu entry

Losing Progress

You won't. GraveStone Mod preserves your items on death. No creeper grief data pack prevents base damage. The pack is designed so you can walk away for a month and come back to exactly the world you left.

Java & Performance

Minecraft 1.21.1 requires **Java 21**. See [MODLIST.md](#) for JDK installation and JVM arguments.

Performance tips:

- ZGC garbage collector with generational mode (`-XX:+UseZGC -XX:+ZGenerational`)
- Distant Horizons handles far rendering — keep vanilla render distance at 10
- ModernFix `dynamic_resources = true` for fastest startup
- If you see missing textures, disable `dynamic_resources` temporarily

Cross-Wave Integration Cheat Sheet

If you want to...	Combine these mods
Auto-build a blueprint	Create Schematicannon + Blueprint and Quill
Power Create from Mekanism	Create Crafts & Additions electric motor/alternator
Bridge TFMG oil to AE2	TFMG Energy Converter + Mekanism universal cable
Store colony resources digitally	AE2 Storage Bus on MineColonies warehouse
Feed colony kitchen automatically	Create conveyor → builder's chest + Farmer's Delight meals
Autocraft Mekanism chemicals	Applied Mekanistics + AE2 Pattern Provider
Cook from your ME network	AE2 Export Bus on Farmer's Delight cooking pot
Get combat buffs before bosses	Farmer's Delight meal + Brewin' And Chewin' drink (effects stack)
Find the best loot	Explorer's Compass → YUNG's structure → loot → Apotheosis Rare+ gear
Build a train network	Steam 'n' Rails tracks → Train Utilities stations → Railways Navigator boards
Fly between Aether islands	Immersive Aircraft airship + Aether portal

If you want to...		Combine these mods	
Defend colony perimeter		Create Big Cannons on walls + Guard Towers	
Explore oceans efficiently		Small Ships cog → Ocean's Delight ingredients → YUNG's Ocean Monuments	
Process space ores		Stellaris desh/ostranium → Mekanism 5x line → AE2 storage	
Build a megabase		Building Wands (structure) + Rechiseled blocks (aesthetic) + Handcrafted furniture (interior)	